

Supplementary Material

Evaluation geometry and the limits of cross-region transfer in pre-fire thermal wildfire prediction

Supplementary material for *Evaluation geometry and the limits of cross-region transfer in pre-fire thermal wildfire prediction*.

Section S1 reports the sensitivity analyses and the detail behind the Results; Section S2 the supporting tables; Section S3 the protocol detail and the limitations; Section S5 the target-label recovery curve; Section S6 the code and Section S7 the data. Tables are numbered S1 to S34, independently of the sections, and supplementary equations are numbered (S1), (S2), and so on. Every value is read from a released output, named by its path in the repository <https://github.com/dryuemco/thermal-twin>; paths beginning `paper/labelfix_rerun/` hold the corrected-label outputs. Values computed under the original Manavgat label are marked “original label” wherever they appear.

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43	Section S1.19 of earlier versions is merged into Section S5, so the numbering skips it; the S4
44	record of earlier versions is replaced by Sections S1.22 and S2.4.

45 **S1 Sensitivity analyses and detail**

46 Each arm below varies one design choice and leaves everything else fixed. None changes a
 47 conclusion in the main text. Section S1.6 is the exception in one respect: it is not a robustness
 48 check but a test of a claim the introduction makes, and the claim is not upheld.

S1.1 Evia AOI and prevalence

The raw transfer arms that involve Evia with Bejís, Manavgat and Muğla were repeated with the legacy, high-prevalence Evia box. Every qualitative conclusion is unchanged. Thermal raw transfer AUCs move by up to 0.134, the largest being Evia to Manavgat at 0.543 with the legacy box against 0.677 with the extended one, and no direction changes side of the chance line (paper/labelfix_rerun/round7/r7f_evia_legacy.csv). On the legacy box the natural-vegetation population is smaller and its burned prevalence higher than the extended box's 0.287 (Table S15).

S1.2 CORAL regularisation

Over nine λ values from 0 to 10^{-1} on four directions (Bejís and Muğla, and Manavgat and Muğla, both ways), CORAL transfer AUC moves by at most 0.014 within any direction and model, and by at most 0.009 for the thermal model (paper/labelfix_rerun/exports/coral_lambda_sensitivity_b74d643e/metrics.csv). No value of λ was selected on performance.

The Manavgat and Bejís pair is not in that sweep, and it is the pair whose adapted result depended on λ under the original label. It was therefore swept separately at $\lambda = 10^{-5}$, 10^{-3} and 10^{-1} (paper/labelfix_rerun/step10/coral_lambda_sensitivity.csv). Bejís to Manavgat gives 0.408 [0.389, 0.427], 0.410 and 0.367, below chance at every λ ; Manavgat to Bejís gives 0.470 [0.444, 0.494], 0.474 and 0.466. No adapted direction of this pair has an interval above chance at any λ , so no conclusion of the paper depends on the choice of λ within this range.

S1.3 Blocking scale

Recomputing the transfer quantities at 10-cell (≈ 5 km) blocking from the re-frozen per-cell predictions widens the intervals and moves the paired-delta verdict counts, from twelve positive, seven negative and one uncertain at 1 km to six, five and nine at 5 km. Coarser blocking therefore removes support from eight verdicts and adds none. Across five bootstrap seeds the 5 km counts run 5 to 6 positive, 4 to 5 negative and 9 to 11 uncertain, so they are not exact (paper/labelfix_rerun/round5/out_official/transfer_ci_blocksize.csv, paper/labelfix_rerun/round6/seed_stability.json).

The point estimates are unchanged. That is an identity rather than a result: the blocking scale is the bootstrap *resampling unit*, and each point estimate is computed once over all target cells. The comparison does establish two things. Every verdict that changes, changes towards “no verdict”, and no direction crosses the chance line under the widened intervals. The counts are the fragile part of this paper; the sign pattern is simply not tested by this variation.

Table 1 note (resampling units). The bootstrap resamples spatial blocks, so an interval’s reliability is bounded by the number of blocks carrying at least one burned cell. Those counts fall from 192 to 843 at 2 cells to **6 to 33** at 20 cells. An interval built on six such blocks has no meaningful coverage, so **the 20-cell row should be read as indicative rather than as an interval**. The 10-cell row is the coarsest blocking this design supports properly: every region there has 16 to 70 positive-carrying blocks, and the increment holds at that scale in all five regions.

S1.4 Predictor-window closure

The predictor window was closed 7 and 14 days earlier in all five regions. The thermal contribution stays positive and bootstrap-supported everywhere, and the direction of the change is region-specific. In Manavgat, on the corrected label, it is +0.065 [+0.059, +0.072] at

the canonical window, +0.082 [+0.075, +0.090] at 7 days and +0.058 [+0.051, +0.065] at 14 days, all on the common cohort of that analysis (paper/labelfix_rerun/exports/window_closure_manavgat_2021/). It strengthens in Bejís, from 0.058 to 0.079 at 14 days, and in Muğla, from 0.115 to 0.128. It is flat in Montiferru. It weakens monotonically in Evia, from 0.156 to 0.149 to 0.135. The four regions other than Manavgat are unaffected by the label correction (paper/labelfix_rerun/exports/window_closure_<region>/tables/thermal_contributions.csv). What holds in every region is survival, not improvement.

S1.5 Quality screening of the coarse thermal input

Two of the five regions' MODIS inputs are quality-screened and three are not. The split follows export date rather than design, and it induces an elevation-correlated change at the input, at $r = +0.615$ in Manavgat. Manavgat's downstream chain was therefore rebuilt from a quality-screened input on the corrected label and compared with the unscreened arm. The current pipeline's step7 refuses the unscreened raster, because it carries no nodata tag and 8.1 % exact zeros, so the unscreened arm runs the step7 of export time and the screened arm the current one: the two differ in code version as well as in screening. The result does not change. Elevation stays at 0.232, no other signed AUC moves by more than 0.005 (downscaled LST, -0.0044), and the within-region increment moves from +0.067 to +0.068 (paper/labelfix_rerun/round6/qc/qc_compare_manavgat_corrected.json; Section S3.5(xii)). The reason is structural: elevation is a DEM variable the screening cannot touch, and fusion falls back on the MODIS-derived surface across only 2.14 percentage points of coverage. The Muğla arm was not re-examined.

S1.6 Normalised against absolute dryness channels

Section 1.2 argues that an internally normalised index should be less exposed to absolute-temperature offsets between regions than raw land surface temperature. The thermal block contains both kinds, so the argument can be tested directly as a feature-set contrast.

The normalised channels are the two that hold their direction in the same-geography two-event comparison of Section S1.13, where the absolute channels move. That is a statement about one region across two fires. Across regions it does not hold: `1st_anomaly_mean` is itself one of the predictors whose reversal is bootstrap-supported on the frames as drawn (Table S10), which is why Section S1.14 drops it alongside elevation. Being internally normalised protects a channel against the offset between two seasons in one place. It does not, on this evidence, protect it against a change of place.

Three feature sets were run over all twenty directions and all five within-region folds, with the classifier, population, folds and bootstrap held fixed. The harness aborts unless its reference configuration lands on the re-frozen exports, and it does exactly: the maximum absolute difference between the reference arm and the step9b transfer AUCs is 0.000000 across all twenty directions. Source for this section and the next two:

`paper/labelfix_rerun/round3/no_coord_channels.json` and

`paper/labelfix_rerun/code/model_capacity.json`, summarised by

`paper/labelfix_rerun/round7/r7e_supplement_tables.py`.

Table S23. Feature-set arms. Mean over the twenty transfer directions and the five regions.

Feature set	Mean transfer AUC	Directions > 0.5	Mean within-region AUC
Baseline only	0.519	13	0.797
Baseline + normalised anomalies	0.524	14	0.862
Baseline + absolute surface state	0.531	13	0.864
All ten features (reference)	0.527	13	0.896

The advantage is not found. Per direction, the normalised set minus the absolute set has a mean of -0.006 . It is positive in 11 of 20 directions and runs from -0.099 to $+0.087$, so the spread is an order of magnitude larger than the difference. Within region the two sub-blocks are indistinguishable, at 0.862 and 0.864, and each recovers most of the gap between the baseline's 0.797 and the full model's 0.896. The comparison is between two sub-blocks of one thermal set on one cohort. It does not test normalised dryness indices in general, nor a normalisation fitted against a pooled multi-region reference.

S1.7 The coordinate-informed channels

downscaled_1st_mean and fused_1st_mean come from a per-region downscaling model whose own inputs include coordinates, the one route by which a coordinate-derived surface re-enters a feature set that excludes coordinates (Section S3.4). A coordinate-smoothed surface is by construction locally informative and non-portable, so if the within-region increment depended on it, that increment would be an artefact of the smoothing rather than a finding about thermal state. Both channels were dropped and everything refitted, over all twenty directions and all five within-region folds.

Table S24. The within-region increment without the two coordinate-informed channels.

Region	Increment, full set	Without the two channels	Retained
Manavgat 2021	+0.067	+0.064	95 %
Bejís 2022	+0.056	+0.046	82 %
Muğla 2021	+0.116	+0.097	84 %
North Evia 2021	+0.153	+0.145	94 %
Montiferru 2021	+0.101	+0.105	103 %
Mean	+0.099	+0.091	92 %

The increment does not depend on them. It stays positive in every region and retains 82 % to 103 % of its size. Mean transfer is likewise unchanged, at 0.528 without the two channels against 0.527 with them and 0.519 for the baseline alone.

S1.8 Model capacity

Every number in the paper comes from one random forest with unlimited depth and `min_samples_leaf = 3`, the configuration most able to encode local structure and least able to extrapolate. Three further estimators were therefore run over the same twenty directions and the same five within-region folds.

Table S25. Four estimators, within region and in transfer.

Estimator	Within-region AUC	Within increment	Transfer AUC	Transfer increment	Above chance
Random forest, depth unlimited, leaf 3	0.896	+0.099	0.527	+0.007	13 of 20
Random forest, depth 6, leaf 50	0.838	+0.053	0.527	−0.011	13 of 20
Random forest, leaf 200	0.814	+0.046	0.522	−0.019	13 of 20
Penalised logistic regression	0.759	+0.046	0.481	−0.021	11 of 20

Regularisation does not rescue transfer. The four estimators land between 0.481 and 0.527, with eleven to thirteen of twenty directions above chance, and the linear model, the one built to extrapolate, transfers worst. Within-region skill falls as capacity is reduced, from 0.896 to 0.759; transfer does not rise to meet it. The thermal block’s contribution to transfer is +0.007 under the canonical forest and negative under all three alternatives. The ordering of the estimators is not claimed, since it rests on point estimates only.

S1.9 The four evaluations of Section 4.3, in full

Section 4.3 reports four evaluations of the same models (main-text Table 2). The per-scar values, the controls and the per-scar ladder of the thermal increment are here.

Table S1. The four evaluations, per scar. Primary natural-vegetation population, 5 km blocking, 2 km collar. Bejís and Manavgat have no row C, because in each the burned area is a single component, so withholding it leaves nothing to train on. Source `paper/labelfix_rerun/code/matched_holdout.json`.

Region	Scar	Cells	Burned fraction	A whole region	B scar area	C scar withheld	D foreign
Manavgat 2021	1	3,965	0.74	0.882	0.743	n/a	0.542
Bejís 2022	1	1,657	0.66	0.824	0.575	n/a	0.589
Muğla 2021	6	1,266	0.72	0.777	0.564	0.595	0.571
Muğla 2021	1	1,244	0.59	0.777	0.654	0.561	0.514
Muğla 2021	10	940	0.68	0.777	0.674	0.617	0.634
Muğla 2021	8	954	0.57	0.777	0.761	0.542	0.623
North Evia 2021	1	3,059	0.87	0.864	0.745	0.465	0.564
Montiferru 2021	1	758	0.58	0.720	0.622	0.584	0.489
Montiferru 2021	5	195	0.34	0.720	0.461	0.458	0.475

S1.9.1 The pool controls

The same predictions were scored on a random sample of region cells drawn at the scar area's own burned fraction. Over the nine scars that gives 0.793 against the region-wide 0.791, so matching prevalence changes nothing, at -0.002 $[-0.005, +0.001]$. ROC-AUC does not depend on class balance at fixed class-conditional distributions, so this is an implementation check rather than a finding. Scoring the predictions on the scar area gives 0.644, a fall of **+0.149 [+0.087, +0.211]** from the prevalence-matched score. That swaps both pools at once, so each was swapped singly, with the cost always defined as the prevalence-matched score minus the swapped score. Replacing only the negatives costs **0.139 [0.098, 0.181]**, 0.046 to 0.250 per scar. Replacing only the positives costs **-0.002 [-0.055, +0.051]** on average, -0.121 to $+0.137$ per scar. The cost is therefore carried by the negative pool. Every negative in a scar collar is fire-adjacent and shares the terrain, land cover and weather of the positives, whereas a region's negatives include its easy far field (`paper/labelfix_rerun/code/pool_decomposition.json`, `prevalence_control.json`).

S1.9.2 The seven scars and the resampling unit

Rows A to D of Table 2 are means over the **same seven scars**, four in Muğla, two in Montiferru and one in Evia. Under the corrected label the 2,151 added Manavgat burns merge into its one existing scar (2,934 of 2,935 cells), which is why the controls above use nine scars and report 0.791 and 0.644 against Table 2's 0.773 and 0.640. The intervals are Student t over the scars. **Row A is a region-level quantity repeated identically across a region's scars, so its interval is pseudo-replicated and should not be read as coverage.** That repetition propagates into $A - B$ and $A - C$. Clustering by region gives **+0.137 [+0.048, +0.226]** for $A - B$ over the three regions, and **+0.160 [+0.090, +0.230]** over all nine scars in five regions (paper/labelfix_rerun/round7/round7_summary.json). $A - C$ clustered by region is +0.266 [-0.022, +0.553], 3.6 times wider than its scar-level interval and spanning zero. **On the region unit only the frame cost is established.**

The patch definition does not drive the result. Row C over a sweep of minimum scar sizes and buffers runs from 0.541 to 0.561 on nine to six scars, and its means at 2, 5 and 10 km buffers are 0.546 (seven scars), 0.533 (seven) and 0.548 (six) (paper/labelfix_rerun/code/scar_definition_sweep.csv, scar_control.json). Row D's spread over sources is large for single scars (Table S4) and small after averaging: 28 of the 36 source-scar combinations enter it, nine of them below chance.

S1.9.3 Controls and the thermal-increment ladder, full specification

Three controls reuse A's predictions, each averaged over 20 draws without replacement. The **prevalence-matched** control draws $|H_K^+|$ cells from P_R and $|H_K^-|$ from $V_R \setminus P_R$. The **negative-pool** control keeps the drawn burned cells but uses the scar's own negatives H_K^- , and the positive-pool control does the converse. A **within-region half-split** (modelled cells cut at the median of a grid axis, both axes and directions, a split discarded when either half is single-

class) completes the set, under the transfer protocol of Section 3.8 unchanged. The per-split positive counts are unequal and bear on the interpretation (Table S2).

The **thermal-increment ladder** of Section 4.3 applies rows A to D to the paired thermal-minus-baseline difference, on the seven scars of Table 2. Table S26 gives it per scar at both blockings. Averaged over the seven scars, the region-wide increment is +0.095 at 5 km blocking and +0.117 at 1 km; scored on the scar area the same blocked predictions give +0.021 and +0.056; leave-one-scar-out gives +0.024 and a foreign-region model +0.008 at both. Blocked minus leave-one-scar-out is -0.004 $[-0.070, +0.063]$ at 5 km and $+0.031$ $[-0.027, +0.090]$ at 1 km. The region-wide minus scar-frame difference at 5 km is $+0.074$ $[-0.009, +0.157]$ scar-level and $[+0.034, +0.146]$ under a region-cluster bootstrap. The within-region half-split gives +0.028, positive in 13 of 18 splits (paper/labelfix_rerun/inference/ladder_summary.json, paper/labelfix_rerun/code/positive_control.json).

Table S26. The thermal increment by scar: region-wide, on the scar frame, withheld and foreign. Thermal minus baseline ROC-AUC. Region-wide and scar-frame values at 5 km blocking, with 1 km in brackets; leave-one-scar-out and foreign values do not depend on the blocking.

Region	Scar	Region-wide	Scar frame, same blocked model	Scar withheld	Foreign
Muğla 2021	6	+0.079 (+0.116)	-0.016 (+0.018)	+0.025	+0.033
Muğla 2021	1	+0.079 (+0.116)	+0.049 (+0.071)	-0.044	+0.014
Muğla 2021	10	+0.079 (+0.116)	+0.063 (+0.107)	+0.045	+0.042
Muğla 2021	8	+0.079 (+0.116)	+0.083 (+0.092)	+0.012	+0.037
North Evia 2021	1	+0.148 (+0.153)	+0.060 (+0.065)	+0.143	-0.038
Montiferru 2021	1	+0.100 (+0.101)	+0.066 (+0.086)	+0.060	-0.008
Montiferru 2021	5	+0.100 (+0.101)	-0.160 (-0.049)	-0.070	-0.023
Mean		+0.095 (+0.117)	+0.021 (+0.056)	+0.024	+0.008

Table S2. Within-region half-split, every split. Source and target positive counts are given because they are unequal, which is the principal limit on this arm: a straight cut does not

231 produce two exchangeable halves. Two splits are unusable because one half of Manavgat
 232 contains no burned cells. Source paper/labelfix_rerun/code/positive_control.json;
 233 checked row by row by paper/code/appendix_tables.py.

Region	Axis	Direction	Source positives	Target positives	Thermal AUC	Baseline AUC
manavgat 2021	east-west	low to high	1,795	1,140	0.761	0.793
manavgat 2021	east-west	high to low	1,140	1,795	0.712	0.638
manavgat 2021	north-south	low to high	0	2,935	n/a	n/a
manavgat 2021	north-south	high to low	2,935	0	n/a	n/a
bejis 2022	east-west	low to high	640	460	0.750	0.784
bejis 2022	east-west	high to low	460	640	0.578	0.560
bejis 2022	north-south	low to high	357	743	0.713	0.662
bejis 2022	north-south	high to low	743	357	0.603	0.613
mugla 2021	east-west	low to high	1,449	1,462	0.448	0.378
mugla 2021	east-west	high to low	1,462	1,449	0.636	0.628
mugla 2021	north-south	low to high	774	2,137	0.533	0.503
mugla 2021	north-south	high to low	2,137	774	0.294	0.549
evia 2021 extended	east-west	low to high	2,052	612	0.589	0.506
evia 2021 extended	east-west	high to low	612	2,052	0.645	0.482
evia 2021 extended	north-south	low to high	2,564	100	0.649	0.513
evia 2021 extended	north-south	high to low	100	2,564	0.533	0.495
montiferru 2021	east-west	low to high	410	129	0.475	0.461
montiferru 2021	east-west	high to low	129	410	0.507	0.513
montiferru 2021	north-south	low to high	438	101	0.583	0.511
montiferru 2021	north-south	high to low	101	438	0.536	0.447

234 **Table S3. Leave-one-scar-out at a 2 km buffer, every scar.** Muğla and Montiferru are the
 235 only regions containing more than one burned component of at least 50 cells. Muğla is the

236 only region where holding one out still leaves the source model properly trained in every arm;
 237 Montiferru's two components are very unequal, so one of its arms retains 472 source positives
 238 and the other 97. Muğla's four arms mean 0.579 and the three arms in the other regions 0.502;
 239 the pooled 0.546 is row C of Table 2. Source
 240 paper/labelfix_rerun/code/scar_control.json; checked row by row by
 241 paper/code/appendix_tables.py.

Region	Component	Source positives left	Target positives	Target cells	AUC
evia 2021 extended	1	11	2,653	3,059	0.465
montiferru 2021	1	97	442	758	0.584
montiferru 2021	5	472	67	195	0.458
mugla 2021	6	1,997	914	1,266	0.595
mugla 2021	1	2,173	738	1,244	0.561
mugla 2021	10	2,272	639	940	0.617
mugla 2021	8	2,363	548	954	0.542

242 The evaluation populations of the arms are not comparable with each other or with the
 243 transfer targets. A held-out scar with its 2 km collar contains only fire-adjacent negatives,
 244 whereas a whole target region includes its easy far field; the burned fractions, 34 to 87 %
 245 against 7.0 to 28.7 %, are a symptom of that rather than the cause.

246 **Table S4. The foreign-region arm, decomposed by source.** Each held-out scar area is
 247 scored with a model fitted on each of the other four regions. Row D of Table 2 is the mean
 248 over the seven scars that carry a row C, 28 of the 36 combinations below; over all nine scars
 249 the mean is 0.556. Source paper/labelfix_rerun/code/d_per_source.json; checked row
 250 by row by paper/code/appendix_tables.py.

Target region	Scar	Mean over sources	Min	Max	Spread
Bejís 2022	1	0.589	0.519	0.631	0.112
North Evia 2021	1	0.564	0.374	0.704	0.329
Manavgat 2021	1	0.542	0.438	0.665	0.227
Montiferru 2021	1	0.489	0.457	0.513	0.056
Montiferru 2021	5	0.475	0.399	0.597	0.197
Muğla 2021	1	0.514	0.410	0.608	0.198
Muğla 2021	6	0.571	0.545	0.597	0.052
Muğla 2021	8	0.623	0.533	0.724	0.192

Target region	Scar	Mean over sources	Min	Max	Spread
Muğla 2021	10	0.634	0.433	0.716	0.283

Over all 36 combinations the mean is 0.556, the range 0.374 to 0.724, and ten fall below chance. The mean spread across the four sources for a single scar is 0.183 over the nine scars and 0.187 over the seven. Averaging over sources is what makes row D comparable with row C, which is fitted on one region; it is not a claim that the choice of foreign source is immaterial.

Table S5. Prevalence is not the cause of the evaluation-area effect. The same fitted model and the same out-of-fold predictions are scored three ways: on the whole region, on a random sample of region cells drawn at the scar area's own burned fraction, and on the scar area. Twenty draws per scar, seed 42. Nine scars, since this control needs no leave-one-scar-out arm. Values at 4 dp, as stored in the source paper/labelfix_rerun/code/prevalence_control.json; checked row by row by paper/code/appendix_tables.py.

Target region	Scar	Burned fraction	A whole region	A' prevalence-matched	B scar area
Manavgat 2021	1	0.74	0.8822	0.8837	0.7434
Bejís 2022	1	0.66	0.8245	0.8251	0.5748
Muğla 2021	6	0.72	0.7773	0.7782	0.5640
Muğla 2021	1	0.59	0.7773	0.7787	0.6540
Muğla 2021	10	0.68	0.7773	0.7835	0.6735
Muğla 2021	8	0.57	0.7773	0.7796	0.7609
North Evia 2021	1	0.87	0.8642	0.8632	0.7451
Montiferru 2021	1	0.58	0.7199	0.7179	0.6215
Montiferru 2021	5	0.34	0.7199	0.7285	0.4611
Mean			0.7911	0.7932	0.6443

A minus A', the effect of prevalence alone, is -0.002 [-0.005 , $+0.001$]. A' minus B, which swaps both pools, is $+0.149$ [$+0.087$, $+0.211$]; Section S1.9.1 attributes it to the negative pool.

S1.10 The transfer-gap decomposition, in full

The recovered fraction of Section 3.10 is at most $+0.28$ among the directions that started below chance (Bejís to Evia), and seven directions have negative recovery, five of them with

intervals entirely below zero; Manavgat to Muğla (−0.03 [−0.06, +0.01]) and Muğla to Bejís (−0.07 [−0.16, +0.01]) are point-estimate cases. Values are from four_aoi_decomposition.csv of the re-frozen outputs (paper/labelfix_rerun/round5/tables/corrected/). Section 4.3 shows that the within-region reference in the denominator is not matched to a transfer evaluation, and Section 4.4 that the raw column depends on the frame, so the fractions are within-protocol quantities.

Table S6. Transfer-gap decomposition (four-AOI set, 12 directions). Within = target's within-region thermal AUC; best adapted = the better of z-score and CORAL, a choice that uses target labels; recovered fraction = (adapted − raw)/(within − raw), signed and unclipped, with paired bootstrap CI (1000 replicates, 2-cell blocks). Montiferru directions are not part of this decomposition. The status column asks whether the *adapted* value clears chance, using the 2-cell adapted intervals of Table S16.

Direction	Within	Raw	Best adapted (method)	Recovered fraction [CI]	Status
Bejís→Evia	0.912	0.383	0.532 (z-score)	+0.28 [+0.23, +0.32]	adapted above chance
Muğla→Manavgat	0.908	0.345	0.485 (z-score)	+0.25 [+0.22, +0.28]	adapted, chance not excluded
Evia→Bejís	0.918	0.448	0.549 (z-score)	+0.22 [+0.16, +0.27]	adapted above chance
Bejís→Manavgat	0.908	0.314	0.406 (CORAL)	+0.15 [+0.13, +0.18]	adapted still below chance
Manavgat→Bejís	0.918	0.396	0.467 (CORAL)	+0.14 [+0.09, +0.17]	adapted still below chance
Manavgat→Muğla	0.859	0.438	0.427 (z-score)	−0.03 [−0.06, +0.01]	negative recovery
Muğla→Bejís	0.918	0.583	0.560 (CORAL)	−0.07 [−0.16, +0.01]	negative recovery
Evia→Muğla	0.859	0.577	0.530 (CORAL)	−0.17 [−0.22, −0.11]	negative recovery
Muğla→Evia	0.912	0.653	0.563 (CORAL)	−0.35 [−0.43, −0.27]	negative recovery
Bejís→Muğla	0.859	0.618	0.518 (z-score)	−0.42 [−0.51, −0.34]	negative recovery
Manavgat→Evia	0.912	0.654	0.529 (z-score)	−0.48 [−0.56, −0.42]	negative recovery
Evia→Manavgat	0.908	0.677	0.417 (CORAL)	−1.13 [−1.27, −1.00]	negative recovery

S1.11 The thermal sign, stratified

Section 4.4 reports that four of five regions agree on a negative association between pre-fire surface temperature and burning, and that Manavgat joins them once elevation is held. The per-region values are here. Signed AUC against burned within the 10 km collar. The stratified

columns pool within-stratum concordance over deciles of the named variable, weighting each stratum by its positive-negative pair count. Sources: paper/labelfix_rerun/code/matched_frame_gap.csv (LST raw, within NDVI, within distance; NDVI columns), recomputable by paper/code/verify_matched_gap.py, and paper/labelfix_rerun/round7/r7b_lst_given_terrain.csv (LST within elevation, and LST detrended on elevation).

Table S27. The LST association, stratified, 10 km collar.

Region	LST raw	LST within elevation	LST detrended on elevation	LST within NDVI	LST within distance	NDVI raw	NDVI within LST
Manavgat	0.522	0.403	0.409	0.556	0.454	0.551	0.550
Bejís	0.405	0.509	0.411	0.486	0.350	0.618	0.574
Muğla	0.332	0.363	0.365	0.404	0.367	0.652	0.547
Evia	0.286	0.327	0.312	0.279	0.319	0.663	0.380
Montiferru	0.376	0.392	0.416	0.368	0.485	0.582	0.405

Three readings follow. First, Manavgat’s raw LST association is above 0.5 and falls well below it once elevation is held, whether by deciles or by removing the region’s own linear dependence of LST on elevation (−5.6 K per km); on the full frame the same holds, 0.665 raw against 0.455 and 0.446. Its hot-burn signal is therefore carried by terrain, and with terrain held it shares the sign of the other four. Second, in the other four regions the sign survives stratification within elevation in three and within greenness in all four, so there it is neither a lapse-rate proxy nor an inverse-greenness proxy; in Bejís it moves to 0.509 within elevation deciles and stays below 0.5 after detrending. Third, the reciprocal adjustment runs one way: NDVI reverses in two regions once temperature is held, while LST reverses in none of the four once greenness is held. Aspect and illumination are not used, so terrain is held only through elevation.

S1.12 The LST anomaly under the difference instrument

Section 4.4 reports two elevation reversals, both involving Manavgat, that meet this paper's per-comparison criterion on the collar. The LST anomaly differs between regions on a weaker instrument: bootstrapping the two regions independently and differencing. These are the three pairs that have opposite-sided point estimates and a difference interval excluding zero, all on 1st_anomaly_mean. Signed AUC within the 10 km collar; the two regions are bootstrapped independently under the 10-cell spatial-block scheme and differenced, 1000 replicates, seed 42.

Table S7. Between-region differences in the signed LST-anomaly association, 10 km collar. Pairs with opposite-sided point estimates and a difference interval excluding zero. Source paper/labelfix_rerun/inference/reversal_family_holm.csv (collar frame); checked row by row by paper/code/appendix_tables.py.

Pair	AUC A	AUC B	Difference	95 % CI
Bejís vs Evia	0.392	0.584	-0.191	[-0.295, -0.083]
Evia vs Montiferru	0.584	0.400	+0.184	[+0.012, +0.321]
Bejís vs Muğla	0.392	0.507	-0.115	[-0.219, -0.002]

Twenty-seven of the ninety feature-by-pair differences on the collar clear zero, against about 4.5 expected at nominal 5 % under the null. Four of the twenty-seven are this feature, three of those have opposite-sided point estimates, and two of the three involve Evia. Under a Holm correction applied with normal-approximation p-values, twelve of the ninety survive; under the stricter intersection-union form, which requires both regions' own associations to be significant, none does (Section S1.20). The nine features carry about two independent thermal signals (Section 3.4), so the ninety comparisons are not independent either. This is a weaker result than a reversal.

S1.13 The same-geography event pair

Section 4.4 states the result and why it does not survive the frame test. The design, the two structural asymmetries that have no analogue in the twenty-direction matrix, and the direction of the bias they impose are here; the protocol is in Section S3.3.

This arm was designed to hold place fixed and vary only the fire, which would have separated regional concept shift from everything that differs between study areas. Muğla burned twice, in 2021 and again eleven months later, on the same grid and through the same processing chain, and the 2022 arm is the 2021 population with the 2021 scar removed: 41,730 rows and 2,911 burned for 2021 against 38,790 rows and 331 burned for 2022. **Positive-carrying 5 km blocks: 70 for the 2021 arm and 11 for the 2022 arm**, and eleven is below the sixteen this design sets as its own floor, so the 2022 intervals are indicative, like the 20-cell row of Table 1. Full per-feature values are in Table S12.

On the frame as drawn, elevation reverses with bootstrap support. In 2021 higher ground burned preferentially, at 0.611 [0.532, 0.690]; in 2022 lower ground did, at 0.296 [0.230, 0.355]. The intervals are disjoint and the difference is -0.317 [-0.414 , -0.220].

The collar removes it. The 2022 arm is one compact scar inside the whole Muğla box, so 93.2 % of its cells lie beyond 10 km of any burned cell (median 43.6 km), against 55.3 % for the 2021 arm, the most extreme far field in the cohort. Under the same 10 km collar the 2021 figure barely moves, 0.611 to 0.606 and still supported, while the 2022 figure moves from 0.297 to 0.565, onto the same side of 0.5 as 2021 with an interval covering chance. Slope reverses on neither frame.

What this arm can and cannot evaluate. The 2022 event's own modelling export is not in this tree. The released script (`paper/code/verify_mugla_collar.py`) reconstructs the arm from the 2021 predictor file with the 2022 burned mask, which is valid for elevation, slope

and land cover (they are identical between the arms) and not for the state-dependent channels. Against the pipeline’s own 2022 export the reconstruction agrees on elevation (0.297 against 0.296) and slope (0.559 against 0.558) and diverges on every seasonal channel, by +0.136 on 1st_anomaly_mean, +0.092 on tvdi_difference_mean, −0.078 on ndvi_mean and −0.025 on current_tvdi_mean. **So only elevation and slope can be given a collar verdict for this arm**, and on the thermal channels it is silent.

Feature	2021, full	2022, full	full verdict	2021, collar	2022, collar	collar verdict
elevation_mean	0.611 [0.529, 0.692]	0.297 [0.229, 0.363]	supported reversal	0.606 [0.525, 0.685]	0.565 [0.450, 0.677]	none
slope_mean	0.637 [0.584, 0.689]	0.559 [0.463, 0.643]	none	0.635 [0.578, 0.692]	0.457 [0.356, 0.548]	none
the state- dependent channels	n/a	n/a	n/a	n/a	n/a	not evaluable here

Source paper/labelfix_rerun/code/mugla_two_event_collar.csv. The two arms are not disjoint samples: they share 38,789 of the 2022 arm’s 38,790 cells, and elevation, slope and land cover are identical across all 73,098 grid cells, so only NDVI and the six thermal channels carry new information between them. And the removal is the target’s own positive class, so in the 2022-to-2021 direction not one target positive is in the source training population while 38,789 of 38,819 target negatives are, and membership of the source training set alone separates the 2021 target’s classes at ROC-AUC 0.9996. **We therefore cannot bound what that asymmetry does to a transfer estimate between the two arms.** The known part of the bias runs the safe way: the removed cells are high, with a median of 563 m, so removing them strips high negatives from the 2022 arm and makes the observed reversal smaller rather than larger.

S1.14 The two interventions, in full

(a) Pooled multi-region training (Fig. 6). Training on the pooled primary populations of the other four regions beats the mean of the four single-source models for one target of five, Evia, at 0.715 [0.668, 0.757] against a mean of 0.569, and falls below that mean for the other four, by 0.009 to 0.057. It also beats the best single source only for Evia (0.654); choosing that best source would use target labels, so it is an oracle benchmark rather than an alternative. In every target the pooled model stays 0.20 to 0.48 AUC below the within-region ceiling. For Manavgat and Bejís it falls below chance, only Manavgat with interval support, at 0.426 [0.369, 0.486]. Outside Evia, aggregation does not recover what single-source transfer loses (paper/labelfix_rerun/code/loro_all.json, paper/labelfix_rerun/round7/r7d_pooled_vs_single.csv).

(b) Removing the direction-reversing features. The two predictors whose signed association reversed between regions with bootstrap support on the frames as drawn under the original label are **elevation_mean** and **1st_anomaly_mean**. The arm keeps them, rather than re-selecting under the corrected label (Table S10), and Section 4.4 shows that the frame decides much of the support, so this arm measures what the removal costs under the original protocol. Retraining without them costs **-0.076** of mean within-region AUC, supported in every region (per-region deltas -0.035, -0.130, -0.073, -0.063 and -0.079 for Manavgat, Bejís, Muğla, Evia and Montiferru, every interval entirely below zero), and changes mean transfer by **+0.014 [-0.028, +0.056]**, which spans zero (source paper/labelfix_rerun/round3/feature_drop_transfer.json, summarised by paper/labelfix_rerun/round7/r7e_supplement_tables.py).

Two qualifications belong with those numbers. First, the debit is mostly not the thermal block's. Dropping elevation alone accounts for -0.056 of it, from 0.896 to 0.840. Dropping

the LST anomaly alone accounts for -0.013 , from 0.896 to 0.883. Roughly three quarters of the cost is therefore the removal of a baseline terrain variable. Second, both figures are post-selection: the two predictors were chosen because they reverse, using the same data on which the costs are then estimated, and no correction is applied.

Table S28. Feature-removal configurations.

Configuration	Mean within-region AUC	Mean transfer AUC
full	0.896	0.527
drop elevation_mean	0.840	0.533
drop lst_anomaly_mean	0.883	0.529
drop both	0.820	0.541

Fig. 7 shows the four configurations together. What this measures is a local cost with no compensating transfer gain. It does not measure an exchange, because the transfer side is a null on both arms: the thermal block's own contribution to transfer is $+0.007$ and removing the reversing predictors returns $+0.014$, both with intervals spanning zero.

S1.15 Signed associations on both frames

Section 4.4 states the collar results; the per-region values and the difference-instrument argument are here. Per-feature values on both frames are in Table S14.

The reversals that meet the per-comparison criterion. On the frames as drawn, Table S10 counts fourteen supported pair-level reversals across seven features, every one but the anomaly pair involving Manavgat. Under the collar, elevation is below 0.5 in Manavgat, at 0.376 [0.300, 0.465], and above it in the other four, individually supported in two, Muğla at 0.606 [0.525, 0.685] and Evia at 0.648 [0.550, 0.740] (paper/labelfix_rerun/code/collar_frame_bootstrap.csv). Those two pairs are the only supported reversals left on the collar, and neither survives the Holm intersection-union correction (Section S1.20). The LST anomaly differs between regions on the weaker difference instrument (Section S1.12).

The sign most regions share. Under the collar, LST lies below 0.5 in four of five regions, at 0.405, 0.332, 0.286 and 0.376 in Bejís, Muğla, Evia and Montiferru: a hotter pre-fire surface is associated with less burning there, and the same holds for TVDI. Manavgat is the exception on the raw association, at 0.522 for LST and 0.527 for TVDI, and joins the other four once elevation is held (Section S1.11). Interval support is not uniform: LST is supported in three of the four and TVDI in two, so “four of five agree” is a statement about point estimates. The absolute thermal channels therefore behave here as static land-surface descriptors rather than as a dryness index, and the two internally differenced channels carry no consistent cross-region direction. Compositing depth was not tested and remains an open alternative.

How many independent reversals there could be. The channels whose reversals vanish on the collar are the ones partly measuring terrain: current LST correlates with elevation at -0.695 to -0.125 across the regions and TVDI at -0.722 to -0.298 . Within the collar `fused_1st_mean` correlates with `current_1st_mean` at 0.99 to 1.00, `downscaled_1st_mean` at 0.97 to 0.99 and `current_tvdi_mean` at 0.87 to 0.98, and the two differenced channels at 0.64 to 0.94. Counts over the nine features therefore count features, not independent quantities; in effective dimensions the thermal block is closer to two.

The thermal block’s paired contribution to transfer, with its interval. The two matrices were differenced direction by direction. The mean is **+0.007**; the individual paired contributions span **-0.148 to $+0.132$** , with **twelve positive and eight negative**. A mean near zero here records cancellation, not consistent absence of effect. The directions are not independent, because each region appears in eight of the twenty, so the interval depends on the resampling unit. All four units the design permits span zero (1000 replicates where resampled; source `paper/labelfix_rerun/code/transfer_delta_ci.json`):

431 **Table S29. The mean paired thermal contribution to transfer, by resampling unit.**

Resampling unit	n	95 % interval on the mean paired contribution
Directions, naive bootstrap	20	[−0.023, +0.037]
Unordered pairs, cluster bootstrap (primary)	10	[−0.020, +0.038]
Unordered pairs, Student <i>t</i> on pair means	10	[−0.028, +0.043]
Regions, leave-one-out jackknife	5	[−0.018, +0.033]

432 Holding out Manavgat, Bejís, Muğla, Evia and Montiferru in turn gives +0.0148, +0.0050,
 433 +0.0072, +0.0010 and +0.0087, so no single region carries the mean or reverses its sign. None
 434 of these units propagates within-direction sampling variability; they resample between
 435 directions only.

436 **In precision terms.** Across the twenty directions the thermal model’s PR-AUC averages
 437 **0.181 against a no-skill baseline of 0.157**; the mean of the twenty per-direction lifts is 1.13.
 438 Seven of the twenty fall below their own no-skill baseline at the point estimate. At 1 km
 439 blocking all seven have intervals entirely below it; at 5 km blocking two do, Bejís to
 440 Manavgat and Muğla to Manavgat
 441 (paper/labelfix_rerun/round7/r7a_pr_auc_10cell.csv). Only one direction, Evia to
 442 Manavgat, exceeds twice its baseline, at 0.321 against 0.143 (Table S11).

443 **The within-region increment on the collar.** Running the baseline arm on the collar as well
 444 (paper/labelfix_rerun/round6/cosine/official-collar_increment_and_cosine.csv),
 445 the thermal increment at 5 km blocking is +0.073, +0.030, +0.088, +0.134 and +0.090 across
 446 the five regions, positive in all five, with a mean of +0.083 against +0.087 on the frame as
 447 drawn. It erodes as the frame tightens further: at a 5 km collar the mean halves to +0.042, still
 448 positive in all five. These are point estimates; the intervals of Table 1 are on the frame as
 449 drawn.

450 **The five areas of interest are not comparable frames.** The share of modelled cells lying
 451 beyond 10 km of any burned cell is 58.5 % in Manavgat, 63.1 % in Bejís, 55.3 % in Muğla,

43.7 % in Evia and **2.1 %** in Montiferru, with median distances of 13.1, 13.5, 11.3, 8.0 and 2.7 km (Table S13). In Manavgat the median elevation of modelled cells rises from 330 m within 5 km of the fire to 995 m at 10 to 20 km and 1,273 m at 20 to 50 km, against 287 m for the burned cells themselves.

S1.16 The contrast pair, in full

Manavgat and Muğla lie in the same country and fire year, 306 km apart by centroid, and their burned cells occupy the most similar environmental envelope of any pair in the matrix; Bejís and Montiferru occupy the least similar (Fig. 8). Per-quantity values for both pairs, on both frames, are in Table S18.

For Manavgat and Muğla, on the frames as drawn seven of nine feature-response directions point opposite ways, and all six features supported in both regions have opposite signs, elevation among them. Under the collar the elevation reversal remains, at 0.376 [0.300, 0.465] in Manavgat against 0.606 [0.525, 0.685] in Muğla. Transfer is below chance in both directions on both frames: 0.438 and 0.345 as drawn, 0.493 and 0.433 under the collar, where Muğla to Manavgat's interval, [0.381, 0.491], excludes chance. The most similar pair is among the weakest in the matrix, but not the extreme: under the collar the weakest direction is Manavgat to Bejís at 0.407 and the strongest Muğla to Evia at 0.728. The niche-overlap and applicability measures of Table S18 are full-frame quantities.

Bejís and Montiferru sit at the opposite extreme. Their burned envelopes barely overlap, and they carry the most dissimilar values on every overlap measure. They transfer above chance in both directions, at the point estimate only: neither direction carries a verdict at 5 km blocking, and the marginal applicability audit was not produced for Montiferru. **At the point estimates, high niche overlap did not guarantee transfer and low overlap did not preclude it.** That rests on two pairs; it is not a correlation across pairs, which Section S1.22 treats.

S1.17 The sensitivity arms, summarised

Eight design choices were varied with everything else held fixed: the Evia AOI and its prevalence, the CORAL regularisation constant, the blocking scale, the closure date of the predictor window, the quality screening of the coarse thermal input, the contrast between the normalised and the absolute dryness channels, the removal of the coordinate-informed channels, and the capacity of the classifier (Sections S1.1 to S1.8). None changes a conclusion of the main text. Two bound how the results should be read.

The secondary population. Section 3.5 defines a secondary population of all valid cells, including cropland. It exists for the within-region arm in Manavgat and Bejís only, so it is a two-region sensitivity rather than a cohort-wide one, and no transfer quantity is defined on it. Across the two regions and the three block sizes the thermal increment runs +0.048 to +0.061 with every bootstrap interval above zero, against +0.045 to +0.067 on the primary population, and the paired difference between populations runs -0.011 to +0.011 with no consistent sign (paper/labelfix_rerun/pipeline/robustness/step8_large_block_primary_all_valid/; Table 1). The increment is therefore not an artefact of excluding cropland, which is what this arm was run to test; it says nothing about the transfer results.

The blocking scale. Coarsening the blocks from 1 km to 5 km moves the paired thermal-minus-baseline delta verdicts from twelve positive, seven negative and one uncertain to six, five and nine, and the verdicts on the thermal arm itself from eleven above, seven below and two uncertain to nine, six and five (Section S1.3). Support is removed and never added. The point estimates are unchanged, which is an identity rather than a result.

S1.18 Distance within a region

One further arm measures how skill decays with distance inside a single region. A model is fitted on one half of a region and applied to the other, and target cells are binned by their

distance from the training cells (paper/code/distance_curve.py; corrected output paper/labelfix_rerun/round3/distance_curve.json, summarised in paper/labelfix_rerun/round7/r7h_distance_curve.csv). Means are unweighted over bins, whose positive counts range from 1 to 1,067, and the bins carry no intervals.

Table S32. Within-region skill by separation from the training cells.

Separation from training cells	Bins	Mean target AUC
0 to 5 km	18	0.709
5 to 10 km	16	0.570
10 to 20 km	11	0.519
20 to 40 km	6	0.499
40 to 80 km	3	0.541
80 to 160 km	1	0.421
cross-region, 306 to 2,802 km	20	0.527

By 10 to 20 km inside a single region the model is close to chance. That bounds how far a surface of this kind can be carried from the cells it was fitted on, and it is consistent with Section 4.3, where withholding a scar and replacing the model with a foreign one cost nothing distinguishable.

It does not follow that the cross-region failure needs no regional mechanism. Once the curve reaches the chance floor, any cross-region mean near 0.5 lies on its continuation by construction, so that comparison could not have come out otherwise. And extrapolating an uninformative model does not produce a reliably reversed ranking: Manavgat to Bejís is below chance with interval support on the frame as drawn and at the 10 km collar.

Four limits bound even the descriptive reading. The two distance ranges do not overlap: within-region separations span 2 to 86 km and cross-region separations start at 306 km. The far bins are thin, so their means should not be read closely and the rise at 40 to 80 km is not evidence of anything. The near bins carry exactly the autocorrelation blocked validation exists to remove, so **0.709 is an upper bound on near-field skill rather than an estimate of it.**

And this design separates distance from crossing a study-area boundary, but not from the land

cover, terrain and fire history that covary with it. The arm contributes a length scale for the within-region decay, not an attribution.

S1.20 The frame test, elaborated

Section 4.4 states these results. Sources: `paper/labelfix_rerun/code/aoi_frame_auc.csv` (signed associations by frame), `paper/labelfix_rerun/code/collar_frame_bootstrap.csv` (their 10-cell intervals), `paper/labelfix_rerun/round5/collar/aoi_frame_transfer.csv` (the transfer matrix by frame) and `paper/labelfix_rerun/code/matched_frame_gap.csv` (the matched reference); code under `paper/code/`.

Construction. Restricting every region to cells within 10 km of any burned cell removes only far-field negatives. Every burned cell is at distance zero and is retained at any radius, so the protection against a flattering radius is the sweep over radii, not the retention of positives. The interval criterion was evaluated at the 10 km collar; at the point estimate the 5 and 10 km radii agree. The collar reduces two regions' cell counts substantially, so part of any loss of support is a loss of power. Features supported in both regions of a direction average 2.3 per direction on the full frame and 2.2 on the collar.

Multiplicity. Each frame carries a family of ninety feature-by-pair comparisons (nine features, ten region pairs). Table S30 gives how many pass each criterion. The per-comparison criterion of Section 3.10 leaves two supported reversals on the collar, both on elevation and both involving Manavgat. Neither survives the Holm intersection-union correction, which adjusts both regions' tests together (adjusted $p = 0.33$ for Manavgat against Evia and 0.79 for Manavgat against Muğla, normal approximation).

Table S30. Reversal family, ninety feature-by-pair comparisons per frame. Source

paper/labelfix_rerun/inference/reversal_family_holm.csv,

reversal_per_region.csv.

Criterion	Full frame	10 km collar
Difference interval excludes zero	39	27
... with opposite-sided point estimates	31	21
Per-comparison criterion (each region's own interval excludes 0.5, opposite sides)	14	2
Holm, normal approximation, difference test	27	12
Holm intersection-union (both regions' tests)	5	0

The diagnostics on the collar. Two of the twenty similarity diagnostics are built from the signed associations themselves and were recomputed on the collar. The sign-agreement fraction over interval-supported features, $\rho = +0.52$ $[-0.27, +0.87]$ on the frames as drawn, takes only the values 0 and 1 across sixteen defined directions on the collar and correlates with collar transfer at $\rho = +0.38$ $[-0.17, +0.85]$. The all-feature cosine, $+0.44$ $[-0.29, +0.80]$ as drawn, correlates with collar transfer at **$+0.57$ $[+0.05, +0.88]$** , the only diagnostic interval in the study that excludes zero (paper/labelfix_rerun/round6/diag_collar/, intervals from paper/labelfix_rerun/round7/r7g_collar_diagnostics.json, which reproduces the as-drawn intervals exactly). Three cautions keep it from being a result: it is one variant among about forty uncorrected diagnostic tests, it rests on twenty directions that share regions, and the collar is defined from the burned cells, so the diagnostic needs target labels twice over. The other eighteen diagnostics were not recomputed on the collar, so their collar correlations are unknown.

Data provenance. A quality-screening rebuild had overwritten two regions' 500 m modelling datasets at the pipeline's canonical path, Manavgat's and Muğla's, after the frozen tables were computed; each replacement differs in `downscaled_1st_mean` and `fused_1st_mean`. The pipeline records a SHA-256 for each region's modelling dataset, and that hash identifies the frozen copy. Every arm of Section 4.4 reads each region through

paper/code/_canonical.py, which verifies the hash on load, and reads Manavgat from the corrected re-freeze (SHA-256 5a5e876c...). The provenance of the collar transfer matrix is in paper/labelfix_rerun/round5/collar/PROVENANCE.md.

Table S8. Cross-region transfer under equalised evaluation frames. Primary natural-vegetation population, thermal model, twenty ordered directions per row. Above/below chance are point counts. The supported counts use a 10-cell (≈ 5 km) spatial-block bootstrap on the target, 1000 replicates, seed 42; Table S16 reports the as-drawn matrix under 2-cell (≈ 1 km) blocking. Per-direction bounds are in paper/labelfix_rerun/round5/collar/aoi_frame_transfer.csv.

Source frame	Target frame	Mean target AUC	Above chance	Below chance	Supported above / below	Paired thermal delta
full	full	0.527	13 of 20	7	9 / 6	+0.007
full	10 km	0.559	14 of 20	6	10 / 2	+0.008
10 km	full	0.546	14 of 20	6	10 / 5	+0.014
10 km	10 km	0.589	15 of 20	5	13 / 2	+0.024
5 km	5 km	0.591	16 of 20	4	11 / 0	+0.017

The baseline control moves with the frame: the static baseline transfers at 0.565 against the thermal model's 0.589 on the 10 km collar, a paired difference of +0.024 rather than +0.007. The control still holds in kind: the static predictor class is not the portable one either. The below-chance count moves from seven to five at the point estimate and from six to two with interval support. The largest movers are Bejís to Evia, 0.383 to 0.602, and Bejís to Manavgat, 0.314 to 0.452; the one large fall is Evia to Manavgat, 0.677 to 0.594.

The matched reference. Recomputing the within-region reference on the same frame and at 5 km blocking (paper/code/verify_matched_gap.py):

Table S33. Within-region reference and transfer on matched frames.

Frame	Within-region (5 km blocking)	Mean transfer	Gap
full rectangle	0.814	0.527	0.287
10 km collar	0.786	0.589	0.197

Frame	Within-region (5 km blocking)	Mean transfer	Gap
5 km collar	0.759	0.591	0.169

Paired by target region, the collar shortfall is **+0.197 [+0.091, +0.303]** (Student *t* over the five target regions; per region +0.081 Montiferru, +0.319 Manavgat, +0.172 Muğla, +0.202 Bejís, +0.211 Evia). The shortfall survives on every matched row, but it is 0.197 at the collar, not the 0.29 the unmatched comparison implies, and it shrinks as the frame approaches the fire. At the 5 km collar the within-region increment itself halves but stays positive in all five regions, at a mean of +0.042 (Manavgat +0.035).

S1.21 The transfer matrix and adaptation, elaborated

Section 4.5 states these results.

Raw transfer. Raw target AUC spans 0.314 to 0.677 (Fig. 4). At 2-cell blocking eleven of 20 directions are above chance with interval support and seven below: both directions between Manavgat and Bejís, both between Manavgat and Muğla, both between Bejís and Evia, and Montiferru to Manavgat. At 10-cell blocking the same points give 9 above, 6 below and 5 uncertain; Manavgat to Muğla loses its support and carries no verdict, and no direction changes side of the chance line. Of the below-chance directions, Manavgat to Bejís (0.407 [0.323, 0.493]) and Muğla to Manavgat (0.433 [0.381, 0.491]) keep interval support on the 10 km collar, though both intervals cover chance at the 5 km collar. The sharpest below-chance direction as drawn is Bejís to Manavgat at 0.314 [0.296, 0.332].

Label-blind adaptation. Under region-wise z-scoring the twenty directions span 0.302 to 0.630 and under CORAL 0.406 to 0.624. The compression is not uniform: under z-scoring Bejís to Manavgat moves further below chance, from 0.314 to 0.302. Taking the better of the two adaptations per direction, 15 of the 20 end closer to chance than they began and 5 end further from it; four of those five involve Montiferru and move upward, while the fifth is Manavgat to Muğla moving downward from 0.438 to 0.427. Under CORAL alone sixteen of

603 twenty end closer (paper/labelfix_rerun/round5/matrix20_official.csv). In the six
604 directions where raw transfer was below chance, the best label-free method recovers at most
605 28 % of the gap to the within-region reference (Bejís to Evia). On the twelve-direction subset
606 for which the decomposition is defined, seven directions show negative recovery (Section
607 S1.10).

608 **Seed stability.** Across five bootstrap seeds every 2-cell verdict is stable. At 10 cells, one level
609 verdict and two paired-delta verdicts are not: the level verdict of Evia to Bejís and the paired-
610 delta verdicts of Manavgat to Muğla and Evia to Montiferru
611 (paper/labelfix_rerun/round6/seed_stability.json). The random-forest seed is fixed,
612 so all direction-level intervals are conditional on one fitted source model.

613 **Jackknife.** Dropping one region at a time moves the mean paired contribution between
614 +0.001 (without Evia) and +0.015 (without Manavgat), and never reverses its sign.

615 **S1.22 The similarity diagnostics**

616 Twenty candidate diagnostics from five families were each rank-correlated with the raw
617 thermal transfer AUC over the ordered directions on which they are defined, under one
618 common pair-based bootstrap (unordered pairs resampled with replacement, both directions
619 carried, 2000 replicates, seed 42). They were defined in the project repository on 8 August
620 2026, with their results under the original label, and were not re-selected after the label
621 correction. Table S17 lists all twenty; the source is
622 paper/labelfix_rerun/round5/out_official/all_diagnostics_vs_transfer.csv, with
623 the original-label values alongside in
624 paper/labelfix_rerun/round5/s6_diagnostics_20.md.

625 **On the frames as drawn, no interpretable diagnostic has an interval excluding zero.** The
626 largest correlations are conditional: the sign-agreement fraction over interval-supported

627 features at $\rho = +0.52$ $[-0.27, +0.87]$ and its cosine at $+0.49$ $[-0.24, +0.87]$. Under the original
628 label these two were the only rows whose intervals excluded zero ($+0.84$ and $+0.81$); the label
629 correction removed that. The marginal family (applicability, dissimilarity, climatic distance,
630 domain classification), the niche-overlap family and the regime family all span zero, and
631 geographic distance does not order the matrix ($\rho = -0.17$ $[-0.87, +0.84]$ on the twelve
632 directions where it is defined). The learned domain classifier separates source from target at
633 $AUC \geq 0.96$ for every pair; it always succeeds, which is why it carries no ordering
634 information. The twentieth measure, the signed-AUC vector Spearman over supported
635 features, is defined on only six directions and its interval is degenerate (upper bound equal to
636 the point estimate), so it is listed and not interpreted.

637 **Equal samples.** The families sit on unequal samples: twelve directions for the marginal,
638 applicability, climatic and geographic rows, eighteen for the supported-conditional rows,
639 twenty for the rest. Every row was therefore recomputed on the common twelve-direction and
640 sixteen-direction subsets. On both, every interpretable row spans zero
641 (`paper/labelfix_rerun/round5/out_official/diagnostics_common_subset.json`).

642 **The collar.** Section S1.20 reports the two diagnostics recomputed on the equalised frame, one
643 of which, the all-feature cosine, orders collar transfer with an interval excluding zero; it is
644 reported, with its cautions, and not relied on.

645 **Limits.** Signed associations need burned labels in both regions, so the conditional family is
646 not available before deployment, while the marginal family, which is, fails. Ten effective
647 region pairs give little power: a moderate ordering would usually be missed, so these results
648 show a failure to demonstrate ordering, not the absence of one (Section S3.5(xiii)). Measures
649 built from interval support are unstable where bounds sit near 0.5 (Section S3.5(viii)).

650 S1.23 Additional robustness analyses

651 The analyses in Table S21 were run after the main results, to test whether the headline
 652 quantities depend on choices a reader might question. None was used to select a reported
 653 configuration.

654 **Table S21. Additional robustness analyses (post hoc).** Sources under
 655 paper/labelfix_rerun/: geometry/ (a1 to a5), inference/ (units, equivalence, ladder) and
 656 round7/.

Analysis	Question	Result
Edge-excluded frame cost (a1)	Is the frame cost carried by the scar's edge cells?	Over nine scars, excluding 0, 1 or 2 cells of the scar edge gives $A - B = 0.147$ [0.086, 0.208], 0.137 [0.069, 0.205] and 0.129 [0.055, 0.204]; with the region as unit 0.160, 0.150 and 0.145, every interval above zero
Placebo collars (a2)	Is the cost a property of any fire-shaped frame?	The scar's real positives scored against the negatives of a same-shaped collar placed away from any burned cell reproduce A ($A - \text{placebo} -0.002$ [-0.056, +0.052], eight scars with placements); the real collar costs +0.155 [+0.108, +0.202] more, so the cost is the fire-adjacent negatives
Label-free target frames (a4)	Does a frame drawn without target labels change transfer?	Trimming the target to the source's per-feature range, or to the source's area of applicability (Meyer and Pebesma 2021), gives means of 0.520 to 0.574, and a 20 km window 0.482; every paired thermal Δ interval spans zero
Metric dependence (a5)	Is the frame cost specific to ROC-AUC?	Region-wide minus scar-frame over nine scars: partial AUC ($\text{FPR} \leq 0.1$) 0.088 [0.048, 0.128]; average precision -0.342 [-0.477, -0.206], higher on the scar frame because its prevalence is higher
Collar within-region increment	Does the within-region increment survive the collar?	+0.083 [+0.037, +0.129] over five regions (Student t), against +0.087 [+0.037, +0.136] as drawn
Equalised Δ by resampling unit	Does the collar Δ exclude zero?	Table S31: two of five computable units exclude zero; the two-way estimators are undefined
Equivalence of the equalised Δ	How large a gain is excluded?	Within ± 0.05 under four of five units; no gain above 0.047 at 90 % (pair cluster); not within ± 0.02 under any unit
Within minus transfer increment	Is the local-versus-portable contrast itself supported?	Full frame +0.079 [+0.014, +0.145]; 10 km collar +0.059 [+0.007, +0.110]; 5 km collar +0.026 [-0.012, +0.063]

Analysis	Question	Result
Reversal multiplicity	Do the collar reversals survive correction?	Table S30: two per-comparison reversals, none after Holm intersection-union
scikit-learn version	Do point estimates depend on the library version?	Under 1.5.2 against 1.9.0, single directions move by up to 0.047 and means by at most 0.003; one support count changes (round7/sklearn152/compare_vs_1_9_0.json)

Table S31. The equalised (10 km collar) paired contribution by resampling unit. 20,000

replicates where resampled; 95 % intervals. Source

paper/labelfix_rerun/inference/units.json.

Unit	95 % interval	Excludes zero
Directions, naive	[−0.001, +0.049]	no
Unordered pairs, cluster bootstrap (primary)	[−0.003, +0.051]	no
Target region, cluster bootstrap	[+0.010, +0.040]	yes
Source region, cluster bootstrap	[+0.010, +0.042]	yes
Pigeonhole (source and target)	[−0.005, +0.060]	no
Two-way cluster (CGM)	undefined (negative variance)	n/a
Dyadic-robust	undefined (negative variance)	n/a
Leave-one-region-out jackknife, t	[−0.017, +0.065]	no

The primary unit is fixed in Section 3.7 as the pair cluster, and on it the equalised contribution spans zero. With five regions a percentile bootstrap over regions has at most 126 distinct resamples, so the two region-clustered intervals are coarse.

S2 Supporting tables

These are the per-region and per-direction numbers the Results sections quote, given in full so that every claim can be checked against the values it rests on rather than against a summary of them.

Table S9. Signed univariate AUC of each predictor against burned, by region. Primary

natural-vegetation population; 10-cell (~5 km) spatial-block bootstrap, 1000 replicates, seed

42. The AUC is never folded to $\max(\text{AUC}, 1 - \text{AUC})$, so a value below 0.5 means lower

values rank burned and is a direction rather than weakness. **Bold** marks a region whose own

671 interval excludes 0.5. Source
672 paper/labelfix_rerun/round5/tables/step9g_multi_aoi_feature_stability.csv (the
673 pipeline's Step9G five-region synthesis, sha256 c864cd7d...), checked row by row by
674 paper/code/appendix_tables.py.

Feature	Manavgat	Bejís	Muğla	Evia	Montiferru
elevation_mean	0.232 [0.179, 0.288]	0.643 [0.558, 0.729]	0.611 [0.532, 0.690]	0.541 [0.448, 0.626]	0.584 [0.395, 0.762]
slope_mean	0.400 [0.340, 0.459]	0.521 [0.439, 0.605]	0.637 [0.582, 0.686]	0.487 [0.418, 0.554]	0.652 [0.506, 0.771]
ndvi_mean	0.564 [0.499, 0.628]	0.559 [0.497, 0.619]	0.662 [0.616, 0.704]	0.639 [0.575, 0.701]	0.586 [0.450, 0.704]
lst_anomaly_mean	0.509 [0.460, 0.560]	0.418 [0.364, 0.480]	0.485 [0.395, 0.566]	0.640 [0.567, 0.710]	0.395 [0.285, 0.535]
current_lst_mean	0.665 [0.608, 0.719]	0.477 [0.401, 0.547]	0.325 [0.271, 0.382]	0.377 [0.301, 0.456]	0.370 [0.248, 0.513]
current_tvdi_mean	0.677 [0.622, 0.733]	0.517 [0.429, 0.595]	0.336 [0.275, 0.398]	0.362 [0.285, 0.442]	0.356 [0.233, 0.499]
tvdi_difference_mean	0.460 [0.409, 0.510]	0.512 [0.443, 0.583]	0.490 [0.396, 0.575]	0.519 [0.444, 0.589]	0.378 [0.282, 0.497]
downscaled_lst_mean	0.683 [0.626, 0.739]	0.484 [0.400, 0.560]	0.307 [0.253, 0.366]	0.377 [0.297, 0.459]	0.365 [0.240, 0.511]
fused_lst_mean	0.666 [0.610, 0.721]	0.481 [0.404, 0.551]	0.325 [0.272, 0.383]	0.376 [0.300, 0.456]	0.370 [0.248, 0.513]

675 **What counts as a reversal.** A pair of regions is called a reversal only when their signed
676 associations point to opposite sides of 0.5 **and each region's own interval excludes 0.5.** That
677 is stricter than requiring the two regions' intervals to be disjoint, and the difference matters:
678 for current_lst_mean between Manavgat and Bejís the two intervals are disjoint, [0.608,
679 0.719] against [0.401, 0.547], but Bejís's own interval includes 0.5, so Bejís has no
680 established direction to reverse from. That pair is a point reversal, not a supported one.

Table S10. The cross-region reversals that meet the stricter criterion. The strict criterion is applied to Table S9's intervals. Difference intervals are the paired 10-cell block bootstrap of paper/code/ems_inference_multiplicity.py (paper/labelfix_rerun/inference/reversal_family_holm.csv, 1000 replicates), which flags the same fourteen pairs. Checked row by row by paper/code/appendix_tables.py.

Feature	Region A	AUC	Region B	AUC	Difference [95 % CI]
elevation_mean	Manavgat	0.232	Bejís	0.643	+0.411 [+0.312, +0.509]
elevation_mean	Manavgat	0.232	Muğla	0.611	+0.379 [+0.274, +0.476]
slope_mean	Manavgat	0.400	Muğla	0.637	+0.237 [+0.158, +0.313]
slope_mean	Manavgat	0.400	Montiferru	0.652	+0.252 [+0.100, +0.389]
lst_anomaly_mean	Bejís	0.418	Evia	0.640	+0.222 [+0.129, +0.316]
current_lst_mean	Muğla	0.325	Manavgat	0.665	+0.340 [+0.260, +0.413]
current_lst_mean	Evia	0.377	Manavgat	0.665	+0.288 [+0.195, +0.384]
current_tvdi_mean	Muğla	0.336	Manavgat	0.677	+0.341 [+0.253, +0.422]
current_tvdi_mean	Evia	0.362	Manavgat	0.677	+0.315 [+0.221, +0.413]
current_tvdi_mean	Montiferru	0.356	Manavgat	0.677	+0.322 [+0.161, +0.453]
downscaled_lst_mean	Muğla	0.307	Manavgat	0.683	+0.376 [+0.295, +0.451]
downscaled_lst_mean	Evia	0.377	Manavgat	0.683	+0.307 [+0.209, +0.405]
fused_lst_mean	Muğla	0.325	Manavgat	0.666	+0.341 [+0.261, +0.413]
fused_lst_mean	Evia	0.376	Manavgat	0.666	+0.291 [+0.198, +0.386]

Fourteen pair-level reversals across **seven** features, thirteen of them involving Manavgat. Under the original label there were three, across two features, elevation and the LST anomaly; the feature-removal arm of Section 3.11 keeps exactly those two and does not re-select. Twenty-six further pairs reverse at the point estimate only, and they are not counted. The conservative criterion costs the paper findings rather than manufacturing them: a difference interval on the pair, the instrument of Section S1.12, would support seventeen further reversals. None of the fourteen is corrected for multiplicity; Table S30 gives the corrected counts.

S2.1 The transfer matrix in precision-recall terms

ROC-AUC is reported throughout the main text for comparability with the susceptibility literature. A susceptibility surface is used as a ranked area budget, so precision-recall is the

operational quantity, and at target prevalences of 7.0 to 28.7 % the two can differ sharply.

Read from the step9b exports of the re-frozen outputs

(paper/labelfix_rerun/round5/tables/corrected/*/step9b_metrics.json).

Table S11. Thermal transfer, PR-AUC against the no-skill baseline. The baseline is the target's burned prevalence. Lift is PR-AUC divided by that baseline; a lift of 1 is no better than random ranking. Ordered by lift. Checked row by row by paper/code/appendix_tables.py. PR-AUC intervals at 2-cell and 10-cell blocking for every direction are in paper/labelfix_rerun/round7/r7a_pr_auc_10cell.csv.

Direction	ROC-AUC	PR-AUC	No-skill	Lift
Evia → Manavgat	0.677	0.321	0.143	2.24
Manavgat → Evia	0.654	0.407	0.287	1.42
Bejís → Montiferru	0.594	0.289	0.212	1.36
Evia → Montiferru	0.647	0.283	0.212	1.34
Bejís → Muğla	0.618	0.093	0.070	1.33
Muğla → Evia	0.653	0.379	0.287	1.32
Montiferru → Bejís	0.548	0.093	0.072	1.28
Montiferru → Muğla	0.619	0.089	0.070	1.27
Evia → Muğla	0.577	0.086	0.070	1.23
Muğla → Bejís	0.583	0.088	0.072	1.22
Manavgat → Montiferru	0.518	0.243	0.212	1.15
Montiferru → Evia	0.586	0.317	0.287	1.11
Muğla → Montiferru	0.531	0.214	0.212	1.01
Manavgat → Muğla	0.438	0.060	0.070	0.85
Evia → Bejís	0.448	0.059	0.072	0.82
Montiferru → Manavgat	0.404	0.113	0.143	0.79
Bejís → Evia	0.383	0.226	0.287	0.79
Manavgat → Bejís	0.396	0.054	0.072	0.75
Muğla → Manavgat	0.345	0.101	0.143	0.70
Bejís → Manavgat	0.314	0.098	0.143	0.68
Mean	0.527	0.181	0.157	1.13

Seven directions fall below their own no-skill baseline, and only one exceeds twice it. The seven are the same seven that are below chance on ROC-AUC, which is what a reversed ranking predicts in either metric. At 1 km blocking all seven have PR-AUC intervals entirely below the baseline; at 5 km blocking two do, Bejís to Manavgat [0.067, 0.132] and Muğla to

Manavgat [0.071, 0.135] against 0.143. Section 4.4 shows that the below-chance count is largely a property of the evaluation frames.

S2.2 The same-geography event pair, in full

Section S1.13 reports this arm; Section 4.4 shows its elevation reversal to be a frame artefact, and Section S1.13 explains why its thermal channels carry no verdict. The per-feature values are kept here because the arm is what motivated the frame test.

Table S12. Signed univariate feature-burned AUC, Muğla 2021 versus 2022. Raw AUC against burned, never folded to $\max(\text{AUC}, 1 - \text{AUC})$; 10-cell (≈ 5 km) spatial-block bootstrap, 1,000 replicates, seed 42 (Section 3.14). Analysis population 41,730 rows / 2,911 burned (2021) and 38,790 rows / 331 burned (2022). **Positive-carrying 5 km blocks: 70 for the 2021 arm and 11 for the 2022 arm.** Table 1’s note sets sixteen as the floor this design supports at that blocking, so the 2022 intervals here fall below the paper’s own standard and are read as indicative, exactly as the 20-cell row of Table 1 is. The 2022 arm is additionally a single compact scar, so its eleven blocks are contiguous. Read from `paper/step9g_raw/.../mugla_2021__mugla_2022_event_relative/step9g_direction_reversal_table.csv`.

Feature	2021 AUC [95 % CI]	2022 AUC [95 % CI]	Reversal
elevation_mean	0.611 [0.532, 0.690]	0.296 [0.230, 0.355]	bootstrap-supported
current_lst_mean	0.325 [0.271, 0.382]	0.515 [0.434, 0.580]	point only
current_tvdi_mean	0.336 [0.275, 0.398]	0.594 [0.475, 0.674]	point only
downscaled_lst_mean	0.307 [0.253, 0.366]	0.508 [0.435, 0.571]	point only
fused_lst_mean	0.325 [0.272, 0.383]	0.519 [0.436, 0.583]	point only
ndvi_mean	0.662 [0.616, 0.704]	0.707 [0.624, 0.777]	none
slope_mean	0.637 [0.582, 0.686]	0.558 [0.468, 0.634]	none
lst_anomaly_mean	0.485 [0.395, 0.566]	0.380 [0.249, 0.502]	none
tvdi_difference_mean	0.490 [0.396, 0.575]	0.397 [0.265, 0.514]	none

S2.3 The evaluation frames, and the signed associations they produce

Section 4.4 states these results; the per-region values are here.

Table S13. Evaluation-frame geometry of the five study regions. Primary natural-vegetation population. Distance is Euclidean to the nearest burned cell on the 500 m grid, at 0.45 km per cell. Corrected Manavgat label. Computed by `paper/code/appendix_tables.py` from the step8a tables (read through `paper/code/_canonical.py`, which verifies each file's sha256), as `paper/code/verify_aoi_frame.py` does.

Region	cells	burned	median distance to burned	share beyond 10 km
Manavgat	20,511	2,935	13.1 km	58.5 %
Bejís	15,190	1,100	13.5 km	63.1 %
Muğla	41,730	2,911	11.3 km	55.3 %
Evia	9,298	2,664	8.0 km	43.7 %
Montiferru	2,544	539	2.7 km	2.1 %

Table S14. Signed univariate AUC, frame as drawn against a 10 km collar. Point estimates; the intervals that decide the reversal question are given in the text below and in `collar_frame_bootstrap.csv` (10-cell blocks, 1000 replicates, seed 42). Signed and never folded to $\max(\text{AUC}, 1 - \text{AUC})$, so a value below 0.5 is a direction, not weakness. The collar drops no burned cells in any region. Full frame from Table S9's source, collar from `paper/labelfix_rerun/code/aoi_frame_auc_frozen_mugla.csv` (the collar arm with Muğla read from its canonical file). Checked row by row by `paper/code/appendix_tables.py`. Manavgat sits on the other side of 0.5 from the other regions on every row, the collar rows included.

Signed univariate AUC	Manavgat	Bejís	Muğla	Evia	Montiferru	straddles 0.5
elevation, full frame (Table S9)	0.232	0.643	0.611	0.541	0.584	yes
elevation, 10 km collar	0.376	0.614	0.606	0.648	0.581	yes
current LST, full frame	0.665	0.477	0.325	0.377	0.370	yes
current LST, 10 km collar	0.522	0.405	0.332	0.286	0.376	yes
current TVDI, full frame	0.677	0.517	0.336	0.362	0.356	yes
current TVDI, 10 km collar	0.527	0.454	0.342	0.250	0.361	yes

Table S15. Region summary: populations and gate outcomes. Counts from each region's Step 8A dataset statistics; gate fractions from each region's burned-landcover gate output. TSG = the primary natural-vegetation population (tree, shrub and grass). The TSG columns

744 use the canonical modelled population, burnable_tree_shrub_grass **and**
745 valid_for_modeling == True, which is the population every model in this paper was fitted
746 and scored on. Counts are computed from the step8a tables and gate fractions read from each
747 region's gate output (paper/labelfix_rerun/round5/tables/corrected/gates/). Checked
748 row by row by paper/code/appendix_tables.py.

Region	Total cells	Valid cells	Burned	Prevalence (all valid)	TSG cells	Burned in TSG	TSG prevalence	Burned natural-veg fraction	Gate verdict
Manavgat 2021	24,150	24,087	3,046	0.126	20,511	2,935	0.143	0.955	pass
Bejís 2022	15,759	15,759	1,103	0.070	15,190	1,100	0.072	0.991	pass
Muğla 2021	73,098	73,045	3,026	0.041	41,730	2,911	0.070	0.958	pass
North Evia 2021 (extended)	22,925	22,906	2,788	0.122	9,298	2,664	0.287	0.945	pass
Montiferru 2021	3,234	3,173	697	0.220	2,544	539	0.212	0.723	pass

749 **Table S16. Cross-region transfer matrix, thermal model, TSG population.** Target ROC-
750 AUC with 2-cell spatial-block bootstrap 95% CIs (1000 replicates). CORAL is applied after
751 region-wise z-scoring ($\lambda = 10^{-5}$). Generated by paper/code/table_b9.py from the re-frozen
752 outputs (paper/labelfix_rerun/round5/matrix20_official.csv).

Direction	Raw	Region-wise z-score	CORAL
Manavgat→Bejís	0.396 [0.373, 0.422]	0.450 [0.423, 0.477]	0.467 [0.441, 0.491]
Bejís→Manavgat	0.314 [0.296, 0.332]	0.302 [0.282, 0.323]	0.406 [0.388, 0.423]
Manavgat→Muğla	0.438 [0.418, 0.456]	0.427 [0.408, 0.444]	0.417 [0.398, 0.436]
Muğla→Manavgat	0.345 [0.331, 0.359]	0.485 [0.468, 0.502]	0.476 [0.460, 0.493]
Manavgat→Evia	0.654 [0.633, 0.676]	0.529 [0.504, 0.553]	0.504 [0.481, 0.528]
Evia→Manavgat	0.677 [0.658, 0.697]	0.404 [0.386, 0.421]	0.417 [0.399, 0.435]
Bejís→Muğla	0.618 [0.601, 0.635]	0.518 [0.501, 0.535]	0.507 [0.489, 0.524]
Muğla→Bejís	0.583 [0.561, 0.607]	0.535 [0.512, 0.557]	0.560 [0.538, 0.581]
Bejís→Evia	0.383 [0.363, 0.402]	0.532 [0.509, 0.551]	0.499 [0.479, 0.518]
Evia→Bejís	0.448 [0.426, 0.470]	0.549 [0.524, 0.575]	0.549 [0.525, 0.573]
Muğla→Evia	0.653 [0.636, 0.671]	0.561 [0.543, 0.580]	0.563 [0.545, 0.582]
Evia→Muğla	0.577 [0.560, 0.593]	0.501 [0.485, 0.518]	0.530 [0.515, 0.546]
Montiferru→Manavgat	0.404 [0.385, 0.422]	0.388 [0.367, 0.408]	0.436 [0.416, 0.456]
Manavgat→Montiferru	0.518 [0.474, 0.563]	0.527 [0.484, 0.568]	0.505 [0.461, 0.547]
Montiferru→Bejís	0.548 [0.521, 0.578]	0.574 [0.552, 0.596]	0.569 [0.548, 0.591]
Bejís→Montiferru	0.594 [0.560, 0.631]	0.550 [0.500, 0.601]	0.574 [0.530, 0.621]

Direction	Raw	Region-wise z-score	CORAL
Montiferru→Muğla	0.619 [0.604, 0.634]	0.576 [0.562, 0.589]	0.565 [0.550, 0.579]
Muğla→Montiferru	0.531 [0.495, 0.568]	0.587 [0.549, 0.624]	0.584 [0.547, 0.623]
Montiferru→Evia	0.586 [0.565, 0.606]	0.630 [0.611, 0.649]	0.624 [0.605, 0.641]
Evia→Montiferru	0.647 [0.608, 0.682]	0.568 [0.528, 0.609]	0.581 [0.539, 0.623]

S2.4 The transferability-diagnostics tables

Table S17. All transferability diagnostics versus raw thermal transfer (20 ordered

directions). Spearman ρ with pair-based bootstrap 95 % CIs. Exp. = expected sign. Rows

with $n = 12$ exist only for the four-AOI subset, because those diagnostics were never

produced for Montiferru. The supported-features conditional rows use the 18 directions with

at least one CI-supported feature. Every row from

paper/labelfix_rerun/round5/s6_diagnostics_20.csv, checked by

paper/code/appendix_tables.py. The twenty diagnostics were defined on 8 August 2026

and not re-selected after the label correction.

Diagnostic	Family	Exp.	n dir	Spearman ρ [95% CI]	CI excludes 0
Agreement fraction, supported features	P(y x) conditional	+	18	+0.52 [−0.27, +0.87]	no
Cosine, supported features	P(y x) conditional	+	18	+0.49 [−0.24, +0.87]	no
Cosine, all 9 features	P(y x) conditional	+	20	+0.44 [−0.29, +0.80]	no
Vector Spearman, all 9	P(y x) conditional	+	20	+0.43 [−0.21, +0.80]	no
Agreement count, all 9	P(y x) conditional	+	20	+0.47 [−0.29, +0.80]	no
Schoener's D, 1-D mean	P(x y=1) niche	+	20	+0.10 [−0.53, +0.63]	no
Warren's I, 1-D mean	P(x y=1) niche	+	20	+0.19 [−0.43, +0.77]	no
Schoener's D, PCA-2D	P(x y=1) niche	+	20	+0.03 [−0.59, +0.63]	no
Warren's I, PCA-2D	P(x y=1) niche	+	20	−0.03 [−0.66, +0.56]	no
Mahalanobis, burned centroids	P(x y=1) niche	−	20	−0.22 [−0.82, +0.42]	no
Domain-classifier AUC	P(ix) marginal	−	20	−0.33 [−0.81, +0.36]	no
Predictor-space mean dissimilarity	P(ix) marginal	−	12	−0.13 [−0.67, +0.37]	no

Diagnostic	Family	Exp.	n dir	Spearman ρ [95% CI]	CI excludes 0
Predictor-space p95 dissimilarity	P(ix) marginal	–	12	–0.15 [–0.68, +0.24]	no
Fraction inside weighted AoA	P(ix) marginal	+	12	+0.21 [–0.48, +0.62]	no
Fraction inside unweighted support	P(ix) marginal	+	12	+0.08 [–0.92, +0.76]	no
Climatic distance	P(ix) marginal	–	12	+0.03 [–0.84, +0.84]	no
Geographic distance	geographic	–	12	–0.17 [–0.87, +0.84]	no
Regime distance, log effective- N	P(y) structure	–	20	+0.11 [–0.67, +0.80]	no
Regime distance, largest share	P(y) structure	–	20	+0.11 [–0.66, +0.77]	no
Vector Spearman, supported (≥ 3 feats) ‡	P(y x) conditional	+	6	+0.96 [+0.85, +0.96]	degenerate interval; not interpreted

Table note. A null row means the diagnostic was **not shown to order transfer** on this design, not that it was shown incapable of ordering it. With ten effective region pairs the power is low, and the intervals are wide enough to admit moderate true correlations in either direction. ‡ The last row is a member of the candidate set fixed in advance, so it stays in the table. It is defined on six directions only, and its interval is degenerate (the upper bound equals the point estimate), so it is not interpreted and is not comparable with the other nineteen.

Under the original label the two supported-feature variants cleared zero (+0.84 and +0.81 over sixteen directions). Both select their predictor subset by whether two regions' bootstrap intervals happen to exclude 0.5, a data-dependent selection made on the same data. Under the corrected label neither clears zero (+0.52 and +0.49 over eighteen directions), and no interpretable row does (Section S3.5(viii) on why measures built from interval support are unstable).

Equal-sample check. The families sit on unequal samples: marginal, applicability, climatic and geographic rows on twelve directions, the supported-conditional rows on eighteen, the rest on twenty. Recomputing every row on the common twelve-direction subset, and again on the sixteen directions where the supported-conditional rows are defined, leaves every

778 interpretable row spanning zero; the supported-conditional rows give +0.41 [−0.42, +0.88]
 779 and +0.46 [−0.56, +0.84] on the common twelve. The published values are reproduced to 4.4
 780 $\times 10^{-5}$. Source:
 781 `paper/labelfix_rerun/round5/out_official/diagnostics_common_subset.json`.
 782 **Table S18. The most and least environmentally similar pairs, on both frames.** Schoener’s
 783 D is computed over burned cells only and is therefore collar-invariant. Transfer values are the
 784 two ordered directions of each pair; ranks are out of the twenty directions on the equalised
 785 frame. Corrected Manavgat label. Schoener’s D , per-feature D and as-drawn transfer are read
 786 from Fig. 8’s source,
 787 `paper/labelfix_rerun/round5/out_official/figure_contrast_pairs.json`; collar
 788 transfer and ranks from Table S8’s source,
 789 `paper/labelfix_rerun/round5/collar/aoi_frame_transfer.csv`; the AoA shares from
 790 `paper/labelfix_rerun/round5/collar/aoa_directed_pair_summary.csv`. Every cell is
 791 asserted against those files at build time (`paper/figures/fig8_contrast_pairs.py`).

	Manavgat and Muğla	Bejís and Montiferru
Schoener’s D , mean 1-D	0.799 (highest)	0.479 (lowest)
per-feature D	0.74 to 0.87	0.23 to 0.77
transfer, frames as drawn	0.438, 0.345	0.594, 0.548
transfer, 10 km collar	0.493, 0.433	0.669, 0.624
rank of 20 on the collar, from the bottom	4th, 2nd	16th, 12th
target cells inside the AoA	0.876, 0.531	n/a

792 **S3 Protocol detail and limitations**

793 These protocols are given here in full rather than in Methods, because each is a specification a
 794 reader needs only when checking the corresponding result, and none is needed to follow the
 795 argument. Sections 3.2 to 3.4, 3.7, 3.9, 3.10, 3.13 and 3.14 state in summary what each does.

S3.1 Burned-area label, the reconstructed analysis grid, and the admissibility gate

Labels come from MODIS MCD64A1 Collection 6.1 (MODIS/061/MCD64A1) retrieved through Google Earth Engine (Gorelick et al. 2017), whose omission and commission characteristics (Boschetti et al. 2019) bound every model here. The product is exported onto the 30 m EPSG:4326 reference grid used by the rest of the pipeline, which duplicates each native ~500 m observation across the 30 m pixels beneath it.

The analysis grid is therefore **reconstructed rather than native**: the 30 m grid is partitioned into non-overlapping blocks of $\text{round}(500 / 30) = 17 \times 17$ pixels, giving a nominal cell edge of 510 m, and each cell is identified by its integer block indices `row_500m` and `col_500m`. This approximates the MODIS sinusoidal cell rather than reproducing it, being anchored to the Landsat/EPSSG:4326 reference grid, and marginal blocks are truncated. The cell is square in degrees but not on the ground, about 510 m north to south and 390 to 407 m east to west, so its area is 0.199 to 0.208 km² against the native MODIS cell's 0.215 km² (463 m edge); every block-size label in this paper is therefore the north-south dimension. The collar step of 0.45 km per cell (Section 3.12) is a single value between the cell's two dimensions, so collar radii are nominal to within about 15 %.

A cell's representative burn date is the **mode** of the positive sub-pixel day-of-year values in the block, tested against the region's label window. Because the exported raster carries no out-of-window positives, in this dataset a single in-window positive sub-pixel makes the cell burned. That dilates each scar by up to one cell at its edge, and it can mark a water cell burned where the product does; the primary population keeps only cells dominated by natural vegetation, which excludes water. The fraction of positive sub-pixels agreeing with the modal date is recorded as `burn_date_pixel_agreement_fraction` but no agreement threshold is

imposed. The label never affects a cell’s eligibility for modelling: unburned, all-no-data and out-of-window cells all remain as the negative class.

Earlier burning is handled unevenly and is stated region by region in Section S3.1.1. The only historical exclusion applied by design removes the 2021 Muğla scar from the 2022 event-relative experiment of Section 3.14.

Before any modelling each region passes a gate that answers one question: of the cells labelled burned, what fraction is dominated by natural vegetation? A region is admitted as a wildfire candidate when that fraction reaches 0.50 and at least 30 burned cells are present, and is rejected as a cropland-dominated control when the cropland fraction reaches 0.50 instead. The gate uses ESA WorldCover classes aggregated to the same reconstructed cells. Its purpose is to separate burned area produced by natural-fuel combustion from burned area produced by post-harvest stubble burning, which MCD64A1 does not distinguish. Verdicts are reported in Section 4.1.

S3.1.1 Pre-label and historical burning, by region

Two kinds of earlier burning can contaminate a label. **Pre-label burning** is burning inside the region’s own predictor window, which the pipeline can remove from the analysis universe. **Historical burning** is burning in the five preceding years, which no region screens for. Table S34 gives both, recomputed from the MCD64A1 export of each region (paper/code/ems_labels_1_prelabel.py, ems_labels_2_history.py; outputs paper/labelfix_rerun/labels/r1_prelabel.json, r2_history.json).

Table S34. Earlier burning by region, natural-vegetation population.

Region	Pre-label cells removed	Pre-label burned cells retained	Historical-burn cells (five years)	... of which burned again in the event
Manavgat 2021	none needed (no pre-label burning)	0	0	0

Region	Pre-label cells removed	Pre-label burned cells retained	Historical-burn cells (five years)	... of which burned again in the event
Bejís 2022	exclusion not applied	48 (49 in all valid cells; 10 burned again)	163	0
Muğla 2021	49	0	194	25
North Evia 2021	16	0	366	88
Montiferru 2021	61	0	58	32

840 Excluding the historical-burn cells lowers the mean paired transfer contribution from +0.007
841 to +0.003 (at 10-cell blocking, 5 positive, 4 negative and 11 uncertain verdicts against 6, 5
842 and 9), and leaves the within-region increment positive with interval support in every region,
843 at +0.062 (no change), +0.055, +0.088, +0.162 and +0.111 for Manavgat, Bejís, Muğla, Evia
844 and Montiferru.

845 **S3.1.2 The Manavgat label correction**

846 **The Manavgat 2021 label is a corrected one.** The label first used for Manavgat was
847 exported on 8 July 2026. That was before the month-alignment fix to the MCD64A1 query
848 (commit 183be42, 11 July), and the export was never renewed. It therefore missed the fire's
849 first four days, 28 to 31 July. The defect was a stale export, not a code error. The corrected
850 label only adds burned cells. No cell burned under the original label becomes unburned, and
851 no predictor or validity flag changes. In the primary population the burned count rises from
852 784 to 2,935. The Manavgat outputs were re-frozen with the upstream pipeline at commit
853 6381f4c, run unchanged except for a one-line patch to the window-closure diagnostic and a
854 runtime override of the Earth Engine project (`thermaltwin`), in Python 3.12.10 with scikit-
855 learn 1.9.0 (pins in `ENVIRONMENT.md`). The correction, the re-freeze and its control arm were
856 carried out by the corresponding author with AI-assisted code (Section 3.13); they were not
857 independently repeated by the author of the pipeline. That pipeline version writes one extra
858 column, a flag for burning in earlier years, and it excludes no Manavgat cell. A control arm
859 re-froze the original label with the same code and environment and reproduced the published

outputs to within 10^{-5} . The exceptions are listed in the manifest: file paths, the column list, and one legacy pair that differs at the level of random-forest thread nondeterminism. Differences between the two arms are therefore attributable to the label alone. The reproduction check of Section 3.13 covers the re-frozen outputs. The manifest, hashes and runner scripts are released under `paper/data/manavgat_2021/refreeze/`.

S3.1.3 Effect on Table 1

The Manavgat rows are computed on the corrected label. It adds burned cells, 784 to 2,935 in the primary population, and changes no predictor (Section 3.2). The rows strengthen rather than weaken. Absolute AUCs rise by 0.04 to 0.11 across the three block sizes, and the increment holds at every scale. The 1 km interval narrows from $[+0.055, +0.079]$ to $[+0.060, +0.073]$, because the region now carries 814 positive-carrying 2-cell blocks rather than 235.

S3.2 Transferability diagnostics versus transfer

Twenty candidate diagnostics from five families are each rank-correlated against the same target quantity, the raw thermal transfer AUC over the twenty ordered directions, under one common pair-based bootstrap. The families are marginal predictor-distribution measures $P(x)$, burned-niche overlap $P(x|y=1)$, fire-regime label-pattern structure $P(y)$, and conditional feature-response direction $P(y|x)$.

The **marginal** family includes area-of-applicability-style dissimilarity in scaled, importance-weighted predictor space (Meyer and Pebesma 2021, 2022; Ludwig et al. 2023), climatic and geographic distance, and a learned domain classifier trained to separate source from target cells. The **niche-overlap** family includes Schoener's D (Schoener 1968) and Warren's I (Warren et al. 2008) over the burned cells of each region. The **regime** family includes burned-patch component counts and effective component counts. The **conditional** family is the sign-agreement index: the fraction of predictors whose signed univariate association points the

same way in both regions, computed over all predictors and over the subset whose associations are individually interval-supported.

Two properties of this design are stated in advance because they bound what it can show. The candidate set is recorded in the project repository on 8 August 2026 (commit d0b20af), in the same commit as its results under the original label, and was not re-selected after the label correction; this is a record, not a pre-registration. And with five regions the effective sample is ten unordered pairs, so nineteen computed variants are evaluated at that power with no family-wise error control claimed; where a diagnostic clears an interval test but not a Bonferroni threshold, both are reported.

S3.3 Same-geography event-to-event comparison (Muğla 2021 versus 2022)

Muğla is the one region for which a second fire event is analysed on the identical AOI and analysis grid, which allows the direction of feature-label associations to be compared with geography held fixed. The 2022 experiment is registered with windows anchored to its own event: a 58-day predictor window closing the day before ignition and a 49-day label window opening on it, matching the 2021 durations exactly.

This is not a clean temporal-transfer design and is not presented as one. The 2022 event ignites about six weeks earlier in the season than the 2021 event, so calendar year and seasonal phase are confounded and no difference can be attributed to the year alone. The registered claim is same-geography event-to-event. The pair is deliberately kept out of the twenty-direction matrix, and that exclusion is enforced in released code rather than merely asserted here.

Two further properties bound the comparison. A historical-burn exclusion applies to the 2022 arm alone, masking every cell that burned in 2021 so the previous year's scar does not enter the following year's analysis: it removes 3,073 cells, of which 2,941 belong to the primary

population, taking it from 41,730 rows to 38,790, a drop of 7.0 %. The two arms therefore share 38,789 of 38,790 cells and three byte-identical static predictors, and in the 2022-to-2021 direction every target positive lies outside the source training population. Section S1.13 states what follows for how these two numbers may be read.

Table S19. Study regions, areas of interest and temporal windows. Bounding boxes are in EPSG:4326, as registered. Window lengths in brackets are inclusive day counts. The baseline years are the four window-symmetric years preceding each predictor window.

Region	Bounding box (lon min, lat min, lon max, lat max)	Predictor window	Label window	Baseline years
Manavgat 2021 (Türkiye)	31.05, 36.72, 31.85, 37.35	2021-06-01 to 2021-07-27 (57 d)	2021-07-28 to 2021-08-31 (35 d)	2017, 2018, 2019, 2020
Bejís 2022 (Spain)	-1.05, 39.68, -0.35, 40.15	2022-06-15 to 2022-08-14 (61 d)	2022-08-15 to 2022-09-30 (47 d)	2018, 2019, 2020, 2021
Muğla 2021 (Türkiye)	27.1, 36.6, 28.9, 37.45	2021-06-01 to 2021-07-28 (58 d)	2021-07-29 to 2021-09-15 (49 d)	2017, 2018, 2019, 2020
North Evia 2021 (Greece)	23.05, 38.55, 23.85, 39.15	2021-06-05 to 2021-08-02 (59 d)	2021-08-03 to 2021-09-30 (59 d)	2017, 2018, 2019, 2020
Montiferru 2021 (Italy)	8.45, 40.05, 8.75, 40.27	2021-05-25 to 2021-07-23 (60 d)	2021-07-24 to 2021-08-31 (39 d)	2017, 2018, 2019, 2020

S3.4 Predictor provenance, compositing and the downscaler

Ten predictors are used, four baseline and six thermal, all summarised per cell over the predictor window, through the chain shown in Fig. 2. All optical and thermal predictors come from Landsat 8 Collection 2 Level-2 (LANDSAT/LC08/C02/T1_L2), quality-screened per pixel from the QA_PIXEL band with the water bit deliberately preserved; the coarse-resolution thermal input is MODIS/061/MOD11A1.

- **NDVI**, the predictor-window median of Landsat surface reflectance, is the baseline's one time-varying member.
- **Elevation** and **slope** come from the Copernicus DEM GLO-30, whose heights are referenced to the EGM2008 geoid (EPSG:3855) in metres according to the product handbook; the pipeline applies no datum conversion.

- **Land cover** is ESA WorldCover v200 (Zanaga et al. 2022), entering as the dominant class code.
- **Current LST**, the predictor-window median of Landsat surface temperature in °C.
- **LST anomaly**, a z-score of the current-window LST median against the four baseline years, set to no-data where the baseline standard deviation is below 1.0 °C or observations are too few.
- **TVDI and TVDI difference**. The Temperature-Vegetation Dryness Index (Sandholt et al. 2002) is computed in the LST-NDVI feature space, with wet and dry edges taken as the 2nd and 98th LST percentiles within each of twenty NDVI bins and the index clamped to [0, 1]. The edges are percentiles of the LST values a given scene contains, so they are fitted per AOI and per window, and a TVDI of 0.5 denotes a different physical moisture state in each region. **TVDI difference** is the raw anomaly of that index against the same four window-symmetric baseline years used for the LST anomaly: `tvdi_difference = current_tvdi - mean(baseline_tvdi)`, in index units rather than standard deviations, which is why it is reported alongside the z-scored channel rather than in place of it.
- **Downscaled LST and fused LST**. A MODIS-to-Landsat downscaling model is trained on the predictor window and applied to the full 30 m grid. The fused product equals observed Landsat LST wherever that is valid, and the downscaled surface only where it is not. Gap-filling therefore never replaces or blends a valid observation. The gap-filled share is 0.11 % to 9.70 % by region. The downscaler's own inputs include coordinates, which is the one route by which a coordinate-derived surface re-enters a feature set from which Section 3.13 excludes coordinates. Section S1.7 reports the increment without these two channels.

The six thermal channels carry about two signals. In the primary population current, fused and downscaled LST correlate at 0.97 to 1.00 in every region, current LST and TVDI at 0.88 to 0.98, and the two anomaly channels at 0.71 to 0.94. Two principal components of the six hold 92 % to 98 % of their variance, and their participation ratio is 1.6 to 1.8 (paper/labelfix_rerun/labels/r4_dimensionality.json).

Table S20. Input data. All retrieved through Google Earth Engine.

Input	Product and version	Native resolution	Use	Reference
Burned area	MODIS MCD64A1 Collection 6.1 (MODIS/061/MCD64A1)	500 m, monthly	label	(Giglio et al. 2018, 2021)
Surface reflectance and temperature	Landsat 8 Collection 2 Level-2 (LANDSAT/LC08/C02/T1_L2), QA_PIXEL screening, water bit preserved	30 m	NDVI, current LST, anomaly, TVDI	(U.S. Geological Survey Earth Resources Observation and Science (EROS) Center 2020)
Coarse LST	MODIS MOD11A1 v061 (MODIS/061/MOD11A1)	1 km, daily	downscaled and fused LST (gap-fill 0.11 % to 9.70 %)	(Wan et al. 2021)
Terrain	Copernicus DEM GLO-30 (EGM2008 heights)	30 m	elevation, slope	(European Space Agency 2022)
Land cover	ESA WorldCover v200 (2021)	10 m	dominant class, gate, population	(Zanaga et al. 2022)

Compositing is the predictor-window median; the anomaly channels reference four window-symmetric baseline years (Table S19). WorldCover 2021 post-dates four of the five fires (Section 3.4).

S3.5 Limitations, in full

Section 5.6 lists the limitations that affect the conclusions. All fourteen are given here.

(i) **No meteorological covariates** enter the models, so we cannot say how local skill and portability behave for a mixed thermal-plus-weather predictor set.

(ii) **The same-geography comparison covers one region only**, and even there year and seasonal phase are confounded. That confound cannot be resolved in this study area, and the reason is specific: the two events sit 42 days apart in median burn day-of-year,

neither year contains a second event at the other's phase, and a calendar-matched arm would carry nine burned cells against this design's gate minimum of thirty. The positive-block count for the 2022 arm is separately below the floor this design sets itself, at eleven against sixteen (Section S1.13). Its 331 burned cells also leave the thermal reversals unresolved at interval level, and the pair holds place fixed but not population. Those two arms were computed with the pipeline's unmodified code and the same pinned environment rather than read from its frozen export.

(iii)**All labels derive from a single burned-area product**, MCD64A1 (Giglio et al. 2018), whose omission and commission characteristics (Boschetti et al. 2019) bound every model evaluated here. Its Mediterranean accuracy has been assessed against reference perimeters (Katagis and Gitas 2022). A second 500 m product, VIIRS VNP64A1 (Giglio 2024), covers both years and was not used as a label sensitivity here.

(iv)**Evia remains the most imbalance-atypical population** even after the AOI extension. Its TSG prevalence is 0.287, the highest in the cohort, against 0.070 to 0.212 in the other four.

(v) **Each region contributes one fire season**, so regional concept shift is confounded with event meteorology, and distinguishing them requires multi-year labels.

(vi)**Point estimates depend on the software version**. Rerunning the frame-transfer matrix under scikit-learn 1.5.2 instead of 1.9.0 moves single-direction AUCs by up to 0.023 for the thermal model and 0.047 for the baseline, and paired differences by up to 0.045; the means move by at most 0.003, and one direction changes interval support on each of the as-drawn and 10 km frames (paper/labelfix_rerun/round7/sklearn152/compare_vs_1_9_0.json). All

reported numbers are fixed to one verified version, and exact reproduction requires the archived environment.

(vii) **Manavgat's atypical transfer is localised, not explained, and the localisation rests on one region and one event.** Manavgat is the weakest target (0.435), carries the largest matched shortfall (+0.319) and has the most outlying univariate profile in the cohort. Two processing candidates have been tested and neither accounts for it: the quality screening of its coarse thermal input moves no signed association by more than 0.005 (Section S1.5), and the evaluation frame leaves its elevation reversal supported (Section 4.4). Holding elevation reverses its raw LST association (Section S1.11). The low-lying burned cells explain part of it, but this rests on one region and one event.

(viii) **The interval-support measures are less stable than the point estimates behind them.** Five to six of the pipeline's interval bounds lie within 0.01 of 0.5, under both labels. A single such flag, Manavgat's NDVI, moved the full-frame supported-feature cosine from $\rho = 0.49$ to 0.70 when two equally valid bootstrap streams disagreed on it. The transfer verdicts are steadier. Across five bootstrap seeds, every verdict at 1 km blocking is stable, and the paired split of twelve positive, seven negative and one uncertain holds on every seed. At 5 km blocking, one level verdict and two paired-delta verdicts change with the seed. Every sentence in this paper that leans on an exact count of supported directions should be read at that precision.

(ix) **The five areas of interest are not comparable frames, and this cohort cannot fully repair it** (Section 4.4). We report the equalised arm alongside the frame-as-drawn arm rather than replacing one with the other, because the collar radius is itself a choice and 5 km and 10 km do not agree exactly (0.591 against 0.589). The deeper limitation is

that the frames were fixed upstream of this work, in repo/, so we can restrict them but not extend them; a region whose rectangle is already fire-scale, Montiferru, cannot be given a far field for symmetry. Nor can their independence from outcomes be fully documented: only Montiferru's box is derived by a rule, Manavgat's is dated before its first gate result, and for Bejís and Muğla the version history cannot show the box fixed before the first gate result (Section 3.1). Any future cohort should fix the frame by an explicit accessible-area rule (Barve et al. 2011) before any predictor is computed, and we treat that as the main design lesson of this paper.

(x) **Other classifiers are compared by point estimate only.** The headline numbers use a random forest with unlimited depth, the configuration most able to encode local structure and least able to extrapolate. Section S1.8 shows the transfer result is not an artefact of that choice: three further estimators, including a penalised linear one, all land between 0.481 and 0.527 and place eleven to thirteen of twenty directions above chance. Those are point estimates without intervals, so the ordering among them is not claimed as a result. Other estimator classes were not tried. Across the four that were tried, the negative result does not depend on the estimator.

(xi) **Burn timing and elevation are confounded in Manavgat.** The cells that burned in the first four days lie at a median of 219 m, against 512 m for the later burns and 1,004 m for unburned cells. The effects of timing and elevation cannot be separated in this data.

(xii) **The quality-screening comparison differs in code version as well as in screening.** The current pipeline's step7 refuses the unscreened MODIS input, because the raster carries no nodata tag and 8.1 % exact zeros. The unscreened arm therefore uses the step7 of export time, and the screened arm the current one. The two agree on

elevation to four decimals and on every other signed AUC to within 0.005, the largest move being downscaled LST at -0.0044 . The within-region increment moves from $+0.067$ to $+0.068$. The confound could therefore have hidden an effect only if two effects had cancelled (Section S1.5).

(xiii) **The diagnostic correlations rest on an effective sample of ten region pairs.**

Five regions give twenty ordered directions, but the two directions of a pair share both regions and are not independent. The diagnostic results of Section S1.22, successes and failures alike, read at that power: with ten pairs a moderate true ordering would usually be missed.

(xiv) **The frame cost rests on few scars.** The full four-row comparison of Section 4.3 is defined on seven scars in three regions, and its scar-level intervals are pseudo-replicated because row A repeats across a region's scars. The region-level estimates ($+0.137$ over three regions, $+0.160$ over five) are the ones to quote.

S3.6 Leakage control and reproducibility, in full

An explicit forbidden-column set is enforced at every model fit. Coordinates (lon, lat, row, col and their normalised forms), every burn-date and label-provenance column, and the agreement fraction are excluded from all feature sets, and the enforcement runs as an assertion rather than a convention. The natural-vegetation mask is used only to define the population, never as a predictor. Spatial blocking prevents a cell from sharing a fold with its own neighbours.

Reproducibility. All randomness uses seed 42 and the bootstrap uses 1000 replicates throughout. The transfer and adaptation analysis runs in an environment separate from the upstream pipeline's. Every within-region model was therefore refitted there and compared against the frozen upstream output, and the independently implemented adaptation was

1062 compared against the pipeline’s own. The within-region comparisons agree exactly and the
 1063 twenty directed CORAL transfers to within 1.3×10^{-8} against the re-frozen outputs (1.6×10^{-7}
 1064 against the frozen ones), under the repository’s pre-existing tolerance of 10^{-6} (Section 3.13).
 1065 All numbers here were produced under scikit-learn 1.9.0 or verified against it; the version
 1066 sensitivity is stated in Section S3.5(vi).

1067 **Sensitivity analyses.** The within-region results are repeated across three spatial-block sizes,
 1068 four classifier capacities and, in two regions, a second population; the transfer results across
 1069 both feature sets, the CORAL sweep, the Evia AOI variant and three evaluation frames;
 1070 where a conclusion depends on one of those choices the dependence is reported rather than
 1071 resolved by choosing the favourable setting (Section S1).

1072 **S3.6.1 Resampling units**

1073 Uncertainty is a spatial-block bootstrap. Let $\beta_1, \dots, \beta_M$ be the blocks of Eq. (1) of the main
 1074 text holding cells of F . Replicate b draws M indices u_{bm} uniformly with replacement:

$$1075 \quad F^{*b} = \bigcup_{m=1}^M \beta_{u_{bm}}, \quad \text{CI}_{95} = [Q_{0.025}\{\theta(F^{*b})\}_b, Q_{0.975}\{\theta(F^{*b})\}_b]. \quad (S1)$$

1076 Here θ is the statistic and Q the 2.5 and 97.5 percentiles over 1000 replicates, seed 42. A
 1077 single-class replicate is discarded. Differences are formed within each replicate, so they are
 1078 paired. The block size of each interval is stated with the result. Because it is blocks that are
 1079 resampled, what bounds an interval’s reliability is the number of blocks carrying at least one
 1080 burned cell, and those counts are reported alongside the intervals. Where a verdict rests on too
 1081 few such blocks it is stated as indicative rather than as an interval.

1082 The twenty transfer directions are not independent, since each region appears in eight. The
 1083 **pair-cluster bootstrap** therefore resamples the ten unordered region pairs, each carrying its
 1084 set π_p of ordered directions, and averages the carried values δ_{st} :

$$\bar{\delta}^{*b} = \frac{\sum_{m=1}^{10} \sum_{(s,t) \in \pi_{u_{bm}}} \delta_{st}}{\sum_{m=1}^{10} |\pi_{u_{bm}}|}. \quad (S2)$$

The interval is the same percentile form, over 1000 replicates for the direction-level intervals of Sections 4.4 and 4.5, and 20,000 for the unit comparison of Table S31. Clustering by target region replaces π_p by the four directions sharing a target. A quantity with one value per held-out scar or target region gets a Student t interval over those n units,

$$\bar{x} \pm t_{0.975, n-1} s_x / \sqrt{n}. \quad (S3)$$

The effective sample is thus ten pairs or five regions for direction-level intervals, and at most seven scars from three regions for scar-level ones.

S3.6.2 Reproduction of the re-frozen outputs

The transfer analysis runs in an environment separate from the upstream pipeline's. The repository's own reproduction check therefore refitted every within-region model and all twenty directed CORAL transfers against the frozen upstream output. For Manavgat that output is the one re-frozen on the corrected label (Section 3.2), produced with the upstream pipeline at commit 6381f4c, run unchanged apart from a one-line patch and a runtime override. The patch lets the window-closure module accept a population column the corrected label adds, and its diff is released with the re-freeze; the override points the Earth Engine calls at the project thermaltwin, which changes infrastructure only. The within-region comparisons agree exactly, and the transfer directions to within 1.3×10^{-8} , under the repository's pre-existing tolerance of 10^{-6} . That 10^{-6} tolerance belongs to the CORAL and within-region reproduction check. The frame-transfer script of Section 4.4 fits its forests in parallel, so its values vary from run to run by up to 2×10^{-6} (1.4×10^{-6} in a rerun on 29 September 2026, paper/labelfix_rerun/round7/), within the 10^{-5} tolerance applied to it,

1107 and are stable at the printed precision. The tolerance that applies if the library version is not
1108 pinned is given in Section S3.5(vi).

1109 **S3.7 Transfer-gap decomposition and the concept-shift diagnostic, in full**

1110 For each direction the gap between the target’s own within-region skill and the raw transfer
1111 result is split in two. One part is what the best label-free adaptation recovers, and the other is
1112 what it does not. The recovered fraction is $(\text{adapted} - \text{raw}) / (\text{within} - \text{raw})$, signed and
1113 unclipped, with its interval from the same paired bootstrap; it bounds what covariate-level
1114 correction can achieve. Section 4.3 shows the remainder should not be read as a conditional
1115 residual, because much of it is incurred inside a single region (Section S1.10).

1116 The mechanism is diagnosed by **signed univariate association**. For each numeric predictor
1117 the raw ROC-AUC of that predictor against burned is computed in each region and never
1118 folded to $\max(\text{AUC}, 1 - \text{AUC})$, so a value below 0.5 is read as a direction rather than as
1119 weakness. A reversal is called bootstrap-supported only when the two regions’ point estimates
1120 fall on opposite sides of 0.5 **and each region’s own interval excludes 0.5**, under the same
1121 10-cell spatial-block bootstrap. That is stricter than requiring the two regions’ intervals to be
1122 disjoint: a feature whose intervals are disjoint but one of which straddles 0.5 has not been
1123 shown to point anywhere in that region, so it is recorded as a point reversal only. Section S2
1124 states the rule again beside the counts, and `conditional_similarity_transfer.json` carries
1125 it as machine-readable metadata.

1126 **S3.7.1 Degenerate replicates**

1127 Replicates with $|G| < 10^{-6}$ are dropped.

1128 S3.8 Label-blind adaptation, full specification

1129 **Region-wise z-score.** Each region's numeric features are standardised using its own statistics,
 1130 source statistics from source data and target statistics from target data, never pooled. For
 1131 feature j in region R ,

$$1132 \quad z_{ij} = \frac{x_{ij} - \mu_j^R}{\sigma_j^R}, \quad \mu_j^R = \frac{1}{n_j^R} \sum_{i \in O_j^R} x_{ij}, \quad \sigma_j^R = \left(\frac{1}{n_j^R} \sum_{i \in O_j^R} (x_{ij} - \mu_j^R)^2 \right)^{1/2}, \quad (S4)$$

1133 where O_j^R holds the n_j^R cells with an observed value (ddof 0). A missing value is first set to
 1134 μ_j^R , so it becomes zero, and $\sigma_j^R < 10^{-12}$ is replaced by 1. Land cover is not transformed.
 1135 Target feature statistics, never target labels, thus enter the adapted arms by design. The
 1136 classifier is refitted on the z-scored source and applied to the z-scored target. This removes
 1137 first- and second-order marginal offsets.

1138 **CORAL after region-wise z-score.** The source covariance is aligned to the target's by the
 1139 standard whitening-recolouring map (Sun et al. 2016). With Z_R the $n_R \times d$ matrix of z-scored
 1140 numeric features, one row per cell,

$$1141 \quad Z_s^{\text{al}} = Z_s (C_s + \lambda I)^{-1/2} (C_t + \lambda I)^{1/2}, \quad C_R = \frac{1}{n_R} \sum_{i=1}^{n_R} (z_i - \bar{z}_R)(z_i - \bar{z}_R)^{\top}, \quad (S5)$$

1142 with $\lambda = 10^{-5}$ (ddof 0). Both means are zero after Eq. (S4), so the general map's mean terms
 1143 vanish. Matrix powers use a symmetric eigendecomposition, eigenvalues floored at 10^{-12} .
 1144 Critically **the transform is applied to the source only**; the target stays at Z_t , and the
 1145 classifier is refitted on Z_s^{al} . Neither variant sees a target label, and both are verified label-blind
 1146 at run time. λ sensitivity was assessed over nine values on four of the twenty directions,
 1147 moving transfer AUC by at most 0.014, and no value of λ was selected on performance; the λ

1148 = 1 of the original CORAL formulation lies outside that sweep, while the value used
1149 throughout remains $\lambda = 10^{-5}$ (Section S1.2).

1150 **S5 Target-label recovery curve**

1151 **S5.1 Purpose**

1152 The main text establishes that label-free alignment does not close the transfer gap. The natural
1153 constructive question is what a small number of target labels buys, since that is the resource
1154 label-free methods cannot substitute for. This analysis prices the failure; it is not a proposed
1155 method and not an operational claim, because the labels it uses come from the event being
1156 predicted. It covers three of the five regions (Manavgat, Bejís and Muğla), all six directions
1157 among them. Source: the pipeline’s few-shot recovery diagnostic on the corrected label,
1158 analysis 7348dfe7..., released in
1159 paper/labelfix_rerun/exports/few_shot_recovery_7348dfe7/ (recovery_curve.csv,
1160 repeat_metrics.csv, report.md); every value below is read from it.

1161 **S5.2 Design**

1162 The unit of labelling effort is one 10-cell (≈ 5 km) spatial block of the target. For a budget of k
1163 blocks, k target blocks are drawn from the training folds of a 5-fold spatially blocked split of
1164 the target, their labels are added to the source training set, and the refitted model is scored on
1165 the held-out target folds. Budgets are 0, 1, 2, 4, 8, 16 and 32 blocks, with ten repeats of the
1166 block draw for every $k > 0$. Budget 0 is raw transfer. The **ceiling** is a target-only model at the
1167 same 10-cell blocking (0.777 for Muğla, 0.824 for Bejís and 0.882 for Manavgat), which
1168 reproduces the pipeline’s own large-block values exactly. The **recovered fraction** is (few-
1169 shot – raw) / (ceiling – raw), the share of the gap between raw transfer and the ceiling that the
1170 labels close; it is signed and unclipped. Intervals are **selection intervals**, the 2.5th and 97.5th

percentiles over the ten block draws; they describe which blocks were drawn and nothing else, and no hypothesis test is made.

S5.3 Result

Table S22. Few-shot recovery of target ROC-AUC, thermal model, natural-vegetation population. Raw = source-only transfer (budget 0); ceiling = target-only model at the same 10-cell blocking. Values are the mean over ten block-selection repeats.

Direction	Raw	1 blk	2 blk	4 blk	8 blk	16 blk	32 blk	Ceiling	Gap closed at 32
Bejís → Manavgat	0.314	0.518	0.565	0.659	0.698	0.750	0.820	0.882	89 %
Muğla → Bejís	0.583	0.578	0.603	0.625	0.666	0.743	0.789	0.824	85 %
Manavgat → Bejís	0.396	0.447	0.456	0.541	0.598	0.690	0.752	0.824	83 %
Muğla → Manavgat	0.345	0.358	0.380	0.407	0.437	0.505	0.624	0.882	52 %
Manavgat → Muğla	0.438	0.447	0.453	0.468	0.490	0.532	0.571	0.777	39 %
Bejís → Muğla	0.618	0.576	0.577	0.578	0.598	0.637	0.666	0.777	30 %

At 32 blocks three of six directions close 83 to 89 % of the gap to the ceiling, two of which started below chance; the other three close 30 to 52 %. Thirty-two blocks carry 2,740 to 2,970 labelled cells, 7 to 20 % of the target’s natural-vegetation population. That is not a modest budget: into Bejís the labelled blocks at 32 carry on average 880 of the region’s 1,100 burned cells, and into Manavgat about 1,800 of 2,935.

Recovery is slow where transfer is worst. Muğla → Manavgat, whose raw transfer sits furthest below its ceiling, is still below 0.5 after 8 labelled blocks and closes 52 % at 32.

Small budgets hurt the direction that already transfers. Bejís → Muğla transfers above chance raw (0.618), and few-shot recalibration helps it least: the curve is below raw at 1, 2, 4 and 8 blocks (0.576 to 0.598), overtakes raw only at 16, and closes 30 % at 32. Where the source model already carries a usable relationship, a small target sample perturbs it before it can replace it.

1189 S5.4 Limits

- 1190 1. **Three regions, six directions.** Evia and Montiferru are absent, so this covers six of
1191 the twenty directions and cannot speak to the five-region scope; six directions cannot
1192 support a general label budget, so none is offered.
- 1193 2. **It requires labelled target cells** from the event being predicted. Nothing here is a
1194 label-free method, and nothing here weakens the negative result about label-free
1195 alignment. Whether labels from earlier fires in the same region would serve is not
1196 tested.
- 1197 3. **The interval is a selection interval, and at the top budgets it narrows because the**
1198 **pool is exhausted, not because the value is well estimated.** Per outer fold the target
1199 holds on average about 12 blocks containing both classes in Bejís, 29 in Manavgat and
1200 48 in Muğla, so at 16 and 32 blocks the draw has nearly exhausted the both-class
1201 blocks and repeats select almost the same set (`direction_budget_feasibility.csv`).
- 1202 4. **The ceiling is the 10-cell target-only value,** not the ≈ 1 km within-region figure of
1203 Table 1, and is correspondingly lower. The recovered fractions are only interpretable
1204 against this matched-blocking ceiling, and on the frames as drawn, which Section 4.4
1205 shows are not comparable across regions.

1206 S6 Code

1207 The analysis code is public at <https://github.com/dryuemco/thermal-twin> under the MIT
 1208 licence; this section is a guide to it, and the repository’s README gives the same
 1209 information. The satellite processing pipeline is a separate public repository,
 1210 <https://github.com/emrehann17/satellite-thermal-digital-twin> (MIT licence), at commit
 1211 6381f4c for the re-frozen Manavgat outputs; the other regions’ outputs were produced at
 1212 earlier commits of that repository and are reproduced by it (Section S3.6.2).

Path	Role
<code>paper/code/</code>	Analysis and verification scripts. <code>_canonical.py</code> loads every modelling dataset, verifies its SHA-256 and asserts the leakage exclusions of Section 3.13 at every fit.
<code>paper/code/appendix_tables.py</code>	Rebuilds Tables S2 to S7 and S9 to S18 row by row from source files whose SHA-256 values are pinned in <code>paper/labelfix_rerun/round5/tables/SOURCES.sha256</code> , and compares them with this document.
<code>paper/figures/</code>	One script per figure. Each asserts every plotted value against the frozen outputs and the manuscript text, and checks its layout.
<code>paper/figures/check_all.py</code>	Runs all figure scripts and <code>appendix_tables.py</code> , restores the committed outputs byte for byte, and exits 0 when everything passes.
<code>paper/tex/</code>	Builds the manuscript and this document from their Markdown sources and checks the port: every number survives, and every cross-reference, citation and supplementary reference resolves.
<code>paper/labelfix_rerun/</code>	The corrected-label outputs from which every corrected number is read: <code>code/</code> , <code>geometry/</code> , <code>inference/</code> , <code>labels/</code> and <code>round3/</code> to <code>round7/</code> hold the analysis outputs, <code>pipeline/</code> the re-frozen pipeline outputs for Manavgat, and <code>exports/</code> the released pipeline diagnostics (few-shot recovery, CORAL λ sweep, window closure), each with its SHA-256 in <code>exports/SHA256SUMS.txt</code> .

1213 All commands run from the repository root. The environment is Python 3.12.10 with NumPy
 1214 2.4.4, pandas 3.0.2 and scikit-learn 1.9.0, pinned in `ENVIRONMENT.md`; point estimates depend
 1215 on the scikit-learn version (Section S3.5(vi)). Fig. 1 additionally needs cartopy; without it
 1216 `check_all.py` reports that figure as skipped. Tables S1, S20 to S34 and the prose values are
 1217 not covered by `appendix_tables.py`; they name their source files in their captions or text.

1218 **S7 Data**

1219 All satellite inputs are public and were retrieved through Google Earth Engine (Table S20).

1220 No proprietary or restricted data were used.

1221 The five modelling datasets the analyses read are in the analysis repository, one per region, as
1222 `paper/data/<region>/step8a_500m_modeling_dataset.parquet`, each verified on load
1223 against the SHA-256 recorded in `paper/code/_canonical.py`. For Bejís, Muğla, North Evia
1224 and Montiferru that hash equals the pipeline’s canonical-input record. For Manavgat the file is
1225 the corrected-label re-freeze (SHA-256 5a5e876c...; Section S3.1.2); the pipeline’s record still
1226 names the original-label table (SHA-256 054a1961...), which is not in the repository and is
1227 regenerated by the pipeline. The frozen numeric outputs every table and figure is built from
1228 are in the same repository, identified by the paths and hashes named in the table captions and
1229 in `SOURCES.sha256`. The data derived from Landsat, MODIS, the Copernicus DEM and ESA
1230 WorldCover remain subject to those products’ terms, which the Declarations reproduce.

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